

Question 1- PJ is a 54-year-old male (height: 180 cm, weight: 85 kg, serum creatinine: 110 $\mu\text{mol/L}$), who takes atenolol for rate control of his atrial fibrillation. Atenolol is 100% renally eliminated and follows first-order kinetics. The volume of distribution is 0.6 L/kg (calculated based on ideal body weight).

What is the expected half-life of atenolol in this patient?

Question 1- Solution: 7.4 hours

1. Determine which weight to use:

Height 180 cm (1 inch = 2.54 cm), so $180/2.54 \approx 71$ inches.

$$\text{IBW}_{\text{male}} = 50 \text{ kg} + 2.3 \times (\text{Height in inches} - 60)$$

$$\text{IBW} = 50 + 2.3 \times (71 - 60) = 50 + 2.3 \times (11) = 75.3 \text{ kg}$$

$\text{IBW} < \text{ABW}$, so we will not use ABW

$$130\% \text{ of IBW} = 1.3 \times 75.3 \text{ kg} = 97.9 \text{ kg}$$

$\text{IBW} < \text{ABW} < 130\% \text{ of IBW}$, so we will use IBW

2. Calculate clearance (Cl):

Drug is 100% renally eliminated, so $\text{Cl} = \text{CrCl}$

$$\text{CrCl}_{\text{male}} = \frac{1.2 \times (140 - \text{Age}) \times \text{Weight (kg)}}{\text{SCr } (\mu\text{mol/L})}$$

$$\text{Cl} = \frac{1.2 \times (140 - 54 \text{ yrs}) \times 75.3 \text{ kg}}{110 \mu\text{mol/L}} = 70.6 \text{ mL/min}$$

$$\text{Convert to L/hr: } 70.6 \frac{\text{mL}}{\text{min}} \times 60 \frac{\text{min}}{\text{hr}} = 4236 \frac{\text{mL}}{\text{h}} \times \frac{1 \text{ L}}{1000 \text{ mL}} = 4.24 \text{ L/h}$$

$$\text{Cl} = 4.24 \text{ L/h}$$

3. Calculate volume of distribution (V_d):

The problem states that V_d is calculated based on IBW.

$$V_d = 0.6 \text{ L/kg} \times 75.3 \text{ kg} = 45.2 \text{ L}$$

4. Calculate elimination rate constant (k_e):

$$k_e = \frac{Cl}{V_d}$$

$$k_e = \frac{4.24 \text{ L/hr}}{45.2 \text{ L}} = 0.0938 \text{ hr}^{-1}$$

5. Calculate half-life ($t_{1/2}$):

$$t_{1/2} = \frac{0.693}{k_e}$$

$$t_{1/2} = \frac{0.693}{0.0938} = 7.4 \text{ hours}$$

Question 2 – LL is a 57-year-old female patient (weight: 60 kg) who receives a rapid IV bolus dose of vancomycin 1000 mg at 08:00. Her bloodwork is drawn at 10:00 and 14:00, revealing a serum concentration of 35 mg/L and 12 mg/L respectively.

What is the volume of distribution (V_d) in liters?

Question 2- Solution: 16.7 L

1. Calculate Elimination Rate (k_e):

Time difference (Δt) is 4 hours (14:00 - 10:00).

$$k_e = \frac{\ln(C_1) - \ln(C_2)}{t_2 - t_1}$$

$$k_e = \frac{\ln(35 \text{ mg/L}) - \ln(12 \text{ mg/L})}{4 \text{ hrs}} = \frac{3.55 - 2.48}{4 \text{ hrs}} = \frac{1.07}{4 \text{ hrs}} = 0.268 \text{ hr}^{-1}$$

2. Calculate Initial Concentration (C_0):

$$C_0 = \frac{C}{e^{-k_e \times t}}$$

$$e^{-k_e \times t} = e^{-0.268 \text{ hr}^{-1} \times 2 \text{ hrs}} = 0.585$$

$$C_0 = \frac{35 \text{ mg/L}}{0.585} = 59.8 \text{ mg/L}$$

3. Calculate Volume (V_d):

$$V_d = \frac{D}{C_0}$$

$$V_d = \frac{1000 \text{ mg}}{59.8 \text{ mg/L}} = 16.7 \text{ L}$$

Question 3- A 51-year-old male patient presents to the hospital with altered mental status, stiff neck and a severe headache. He is later diagnosed with meningitis and prescribed ceftriaxone 2 g IV q12h. The clearance is 66.7 mL/min and the volume of distribution is 50L.

Calculate the accumulation ratio (r) expected at steady-state for ceftriaxone.

Question 3- Solution: 1.62

1. Convert clearance to L/hr:

$$66.7 \text{ mL/min} \times \frac{60 \text{ min}}{1 \text{ hr}} \times \frac{1 \text{ L}}{1000 \text{ mL}} \approx 4 \text{ L/hr}$$

2. Calculate elimination rate constant (k_e):

$$k_e = \frac{Cl}{V_d}$$

$$k_e = \frac{4 \text{ L/hr}}{50 \text{ L}} = 0.08 \text{ hr}^{-1}$$

3. Calculate accumulation ratio (r):

$$r = \frac{1}{1 - e^{-k_e \tau}}$$

First, calculate the exponent factor ($e^{-k_e \tau}$):

$$e^{-(0.08 \times 12)} = e^{-0.96} \approx 0.383$$

Then, perform the final calculation:

$$r = \frac{1}{1 - 0.383} = \frac{1}{0.617} = 1.62$$

Question 4- MS is a 42-year-old male patient who has been prescribed propranolol for the management of portal hypertension. To optimize the dose, the clinician needs to estimate the drug's hepatic clearance. MS's cardiac output is measured to be 6 L/min. Hepatic blood flow is 25% of cardiac output. His hepatic extraction ratio (E) is 0.8.

What is the hepatic clearance in L/min?

Question 4- Solution: 1.2 L/min

1. Calculate the hepatic blood flow (Q):

$$Q = 0.25 \times 6 \text{ L/min} = 1.5 \text{ L/min}$$

2. Calculate the hepatic clearance (Cl_h):

$$Cl_h = Q \times E$$

$$Cl_h = 1.5 \text{ L/min} \times 0.8 = 1.2 \text{ L/min.}$$

Question 5- You are working at a pharmaceutical company and have been tasked with preparing the drug monograph for a new oral drug (drug Z). The intravenous formulation of this drug has been on the market for several years, and you would like to determine the bioavailability of the oral formulation. Based on the data listed below, what is the absolute bioavailability (F) of drug Z?

Trial Parameters:

- Drug Z IV Arm: Dose (D_{IV}) = 200 mg, $AUC_{IV} = 40 \text{ mg} \cdot \text{hr/L}$
- Drug Z Oral Arm: Dose (D_{oral}) = 500 mg, $AUC_{oral} = 60 \text{ mg} \cdot \text{hr/L}$

Question 5- Solution: 60%

1. Calculate bioavailability:

$$F = \frac{AUC_{oral}/D_{oral}}{AUC_{IV}/D_{IV}} \times 100$$

$$F = \frac{\frac{60 \text{ mg} \cdot \text{hr/L}}{500 \text{ mg}}}{\frac{40 \text{ mg} \cdot \text{hr/L}}{200 \text{ mg}}} \times 100$$

$$F = \frac{0.12}{0.2} \times 100 = 0.6 \times 100 = 60\%$$

Question 6- A 62-year-old male patient (weight: 70 kg) is admitted to the cardiac care unit (CCU). He is given a loading dose of amiodarone to achieve a target concentration of 15 mg/L. The volume of distribution is 0.5 L/kg and the clearance is 0.04 L/kg/hr.

Calculate the concentration exactly 4 hours after the loading dose is administered (assuming instantaneous distribution).

Question 6- Solution: 10.9 mg/L

1. Calculate Cl:

$$Cl = \frac{0.04 \frac{L}{kg}}{hr} \times 70 \text{ kg} = 2.8 \text{ L/hr}$$

2. Calculate Vd:

$$Vd = 0.5 \frac{L}{kg} \times 70 \text{ kg} = 35 \text{ L}$$

3. Calculate k:

$$k = \frac{Cl}{Vd} = \frac{2.8 \text{ L/hr}}{35 \text{ L}} = 0.08 \text{ hr}^{-1}$$

4. Identify C_0 :

The patient is loaded to a target a concentration of 15 mg/L and the question states to assume instantaneous distribution, so we can assume $C_0 = 15 \text{ mg/L}$ immediately after loading.

5. Calculate the plasma concentration of amiodarone at 4 hours:

$$C_t = C_0 \times e^{-k_e t}$$

$$C_{4 \text{ hrs}} = 15 \text{ mg/L} \times e^{-(0.08 \text{ hr}^{-1} \times 4 \text{ hrs})} = 10.9 \text{ mg/L}$$

Question 7 – Suppose that labetalol has a half-life of 8 hours. A patient presents to the emergency department with bradycardia, hypotension and altered mental status. It is determined that the patient has ingested a toxic quantity of labetalol tablets. Upon assessment, the patient's plasma drug level is found to be 120 mg/L. The target safe level is 5 mg/L.

Calculate how many hours it will take for the drug levels to drop to 5 mg/L.

Question 7- Solution: 36.7 hours

1. Calculate k_e :

$$k_e = \frac{0.693}{t_{1/2}}$$

$$k_e = \frac{0.693}{8 \text{ hrs}} = 0.0866 \text{ hr}^{-1}$$

2. Calculate time (t):

$$C_t = C_0 \times e^{-k_e t}$$

$$\frac{C_t}{C_0} = e^{-k_e t}$$

Solve for time (t).

$$\ln\left(\frac{C_t}{C_0}\right) = -k_e \times t$$

$$t = \frac{\ln\left(\frac{C_t}{C_0}\right)}{-k_e}$$

$$t = \frac{\ln\left(\frac{5 \text{ mg/L}}{120 \text{ mg/L}}\right)}{-0.0866 \text{ hr}^{-1}} = 36.7 \text{ hrs}$$

Question 8 - JR is a 3-year-old child (weight: 15 kg) who has been diagnosed with acute otitis media. The pediatrician prescribes amoxicillin/clavulanic acid 45 mg/kg/day of the amoxicillin component, to be administered in two divided doses. To minimize the risk of antibiotic-associated diarrhea, the pharmacist must select the suspension concentration that results in the lowest exposure to clavulanate.

Available Concentrations:

- Option A: Amoxicillin 400 mg/clavulanic acid 57 mg per 5 mL (7:1 ratio)
- Option B: Amoxicillin 250 mg/clavulanic acid 62.5 mg per 5 mL (4:1 ratio)

Determine the most appropriate concentration and the most appropriate dosing regimen.

Question 8- Solution: Amoxicillin 400 mg/clavulanic acid 57 mg per 5 mL give 4.2 mL PO BID

1. Calculate the total daily amoxicillin requirement:

$$45 \text{ mg/kg/day} \times 15 \text{ kg} = 675 \text{ mg/day}$$

2. Calculate the amoxicillin dose per administration:

$$\frac{675 \text{ mg/day}}{2 \text{ doses/day}} = 337.5 \text{ mg/dose}$$

3. Calculate the clavulanic acid content per dose for option A:

$$\frac{400 \text{ mg amoxicillin}}{5 \text{ mL suspension}} = 80 \frac{\text{mg}}{\text{mL}}$$

$$\frac{337.5 \text{ mg}}{\text{dose}} = 4.2 \text{ mL}$$

$$\frac{57 \text{ mg clavulin}}{5 \text{ mL suspension}} = 11.4 \frac{\text{mg}}{\text{mL}} \times 4.2 \text{ mL} = 47.9 \text{ mg clavulin per dose}$$

4. Calculate the clavulanic acid content per dose for option B:

$$\frac{250 \text{ mg amoxicillin}}{5 \text{ mL suspension}} = 50 \frac{\text{mg}}{\text{mL}}$$

$$\frac{337.5 \text{ mg}}{50 \text{ mg/mL}} = 6.75 \text{ mL}$$

$$\frac{62.5 \text{ mg clavulin}}{5 \text{ mL suspension}} = 12.5 \frac{\text{mg}}{\text{mL}} \times 6.75 \text{ mL} = 84.4 \text{ mg clavulin per dose}$$

5. Determine the most appropriate concentration:

Option A provides the least amount of clavulanic acid per dose. As such, this is the most appropriate choice.

Question 9 - MS is a 68-year-old female (weight: 132 lbs, SCr: 0.9 mg/dL) who is taking clarithromycin 250 mg PO Q12H for severe community-acquired pneumonia. Drug clearance is approximately 80% of her creatinine clearance (CrCl).

Calculate the average steady-state concentration ($C_{p,av,ss}$) in mg/L, assuming 100% bioavailability (F).

Question 9- Solution: 7.66 mg/L

1. Calculate CrCl:

$$CrCl_{\text{female}} = \frac{(140 - \text{Age}) \times \text{Weight (kg)}}{72 \times SCr \text{ (mg/dL)}} \times 0.85$$

$$\text{Convert weight to kg: } \frac{132 \text{ lbs}}{2.2 \text{ lbs/kg}} = 60 \text{ kg}$$

$$CrCl = \frac{(140 - 68) \times 60 \text{ kg}}{72 \times 0.9 \text{ mg/dL}} \times 0.85 = 56.7 \text{ mL/min}$$

2. Convert CrCl to L/h:

$$\frac{56.7 \text{ mL}}{\text{min}} \times \frac{60 \text{ min}}{\text{hr}} \times \frac{1 \text{ L}}{1000 \text{ mL}} = 3.40 \text{ L/hr}$$

3. Calculate drug clearance:

$$Cl_{\text{drug}} = 0.8 \times 3.40 \frac{\text{L}}{\text{hr}} = 2.72 \frac{\text{L}}{\text{hr}}$$

4. Calculate $C_{p,av,ss}$:

$$C_{p,av,ss} = \frac{F \times D}{Cl_{drug} \times \tau}$$

$$C_{p,av,ss} = \frac{1 \times 250 \text{ mg}}{2.72 \frac{\text{L}}{\text{hr}} \times 12 \text{ hrs}} = \frac{250 \text{ mg}}{32.64 \text{ L}} = 7.66 \text{ mg/L}$$

Question 10 - A 72-year-old male patient (weight: 75 kg) has been diagnosed with infective endocarditis. He receives a single 500 mg IV dose of vancomycin. The half-life of vancomycin is 9.5 hours and its volume of distribution is 0.4 L/kg.

Calculate the total Area Under the Curve ($AUC_{0-\infty}$) in $\text{mg} \cdot \text{hr/L}$.

Question 10- Solution: 228 $\text{mg} \cdot \text{hr/L}$

1. Calculate Elimination Rate Constant (k_e):

$$k_e = \frac{0.693}{t_{1/2}}$$

$$k_e = \frac{0.693}{9.5 \text{ hrs}} = 0.073 \text{ hr}^{-1}$$

2. Calculate Total V_d :

$$V_d = 0.4 \text{ L/kg} \times 75 \text{ kg} = 30 \text{ L}$$

3. Calculate Clearance:

$$Cl = V_d \times k_e$$

$$Cl = 30 \text{ L} \times 0.073 \text{ hr}^{-1} = 2.19 \text{ L/hr}$$

4. Calculate AUC:

$$AUC = \frac{\text{Dose}}{Cl}$$

$$AUC = \frac{500 \text{ mg}}{2.19 \text{ L/hr}} = 228 \text{ mg} \cdot \text{hr/L}$$

Question 11 - A pharmacist is tasked with compounding 450 grams of a 12% gel using two available stock strengths of Benzagel® (benzoyl peroxide): a 20% concentrate and a 5% cream.

Calculate the amount (in grams) of the 20% concentrate required to complete the prescription.

Question 11- Solution: 210 g

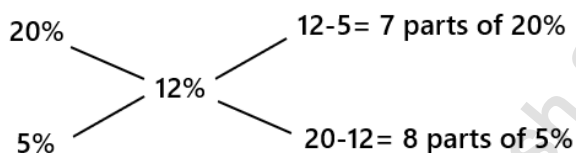
1. Alligation method:

Place the higher concentration (20%) at the top left.

Place the lower concentration (5%) at the bottom left.

Place the desired concentration (12%) in the center.

Subtract diagonally to find the parts:



2. Calculate Total Parts:

$$7 \text{ parts} + 8 \text{ parts} = 15 \text{ parts}$$

3. Calculate Grams of 20% Strength:

$$\frac{450 \text{ g}}{15 \text{ total parts}} = 30 \text{ g/part}$$

$$30 \text{ g/part} \times 7 \text{ parts} = 210 \text{ g of 20\% concentrate}$$

Question 12 – JM is a male patient (weight: 120 kg, height: 6 feet and 2 inches) who is being treated with gentamicin for a severe skin infection. A 1000 mg dose results in a peak concentration of 25 mg/L.

Calculate the Volume of Distribution per kg of Ideal Body Weight (V_d/kg_{IBW}).

Question 12- Solution: 0.49 L/kg (IBW)

1. Calculate Total V_d :

$$V_d = \frac{D}{C_0}$$

$$V_d = \frac{1000 \text{ mg}}{25 \text{ mg/L}} = 40 \text{ L}$$

2. Calculate IBW:

Height 6'2" = 14 inches over 5 feet

$$IBW_{\text{male}} = 50 \text{ kg} + 2.3(\text{in} > 5\text{ft})$$

$$IBW = 50 + 2.3(14) = 82.2 \text{ kg}$$

3. Normalize to IBW:

$$V_{d(\text{normalized})} = \frac{V_{d(\text{total})}}{IBW}$$

$$\frac{40 \text{ L}}{82.2 \text{ kg}} = 0.49 \text{ L/kg}$$

Question 13 – AR is a 78-year-old male (weight: 176 lbs) who has been admitted to the intensive care unit (ICU) due to ventricular tachycardia. He is on a lidocaine infusion at 2 mg/min. Its clearance is 0.6 L/kg/hr.

Calculate the expected steady-state concentration in mg/L.

Question 13- Solution: 2.5 mg/L

1. Convert infusion rate (R_0) to mg/hr:
 $2 \text{ mg/min} \times 60 \text{ min/hr} = 120 \text{ mg/hr}$
2. Convert weight to kg:
 $176 \frac{\text{lbs}}{2.2 \text{ lbs/kg}} = 80 \text{ kg}$
3. Calculate Total Clearance:
 $0.6 \text{ L/kg/hr} \times 80 \text{ kg} = 48 \text{ L/hr}$

4. Calculate C_{ss} :

$$C_{ss} = \frac{R_0}{Cl_{total}}$$

$$C_{ss} = \frac{120 \text{ mg/hr}}{48 \text{ L/hr}} = 2.5 \text{ mg/L}$$

Question 14 - A 63-year-old male patient (weight: 83 kg) is receiving a loading dose of Dilantin® (phenytoin) IV injection. The salt factor of this product is 0.92 and its volume of distribution is 0.65 L/kg. The medical team wishes to target a plasma concentration of 18 mg/L.

Calculate the IV loading dose required to achieve this steady-state concentration (in mg of phenytoin sodium).

Question 14- Solution: 1055.5 mg

1. Calculate Total Volume of Distribution (V_d):

$$V_{d(\text{total})} = 0.65 \text{ L/kg} \times 83 \text{ kg} = 53.95 \text{ L}$$

2. Calculate Loading Dose (DL):

$$D_L = \frac{C_{p_{ss}} \times V_d}{S \times F}$$

$$DL = \frac{18 \text{ mg/L} \times 53.95 \text{ L}}{0.92 \times 1} = 1055.5 \text{ mg}$$

Question 15 - A patient is prescribed prednisone for the management of an acute inflammatory condition. The patient is instructed to take 15 mg once daily for 7 days, then reduce the daily dose by 5 mg every week. Once the patient has been taking 5 mg PO daily for one week, they should continue by reducing the dose by 2.5 mg every week until finished. You have 5 mg tablets of prednisone available in your community pharmacy.

Calculate the total number of tablets to dispense.

Question 15- Solution: 46 tablets

1. Calculate week 1:

$$3 \text{ tabs/day} \times 7 \text{ days} = 21 \text{ tabs}$$

2. Calculate week 2:

$$2 \text{ tabs/day} \times 7 \text{ days} = 14 \text{ tabs}$$

3. Calculate week 3:

$$1 \text{ tab/day} \times 7 \text{ days} = 7 \text{ tabs}$$

4. Calculate week 4:

$$\frac{1}{2} \text{ tab/day} \times 7 \text{ days} = 3.5 \text{ tabs}$$

5. Add up the total number of tablets:

$$21 + 14 + 7 + 3.5 = 45.5 \text{ tablets}$$

6. Round up to the nearest whole number of tablets:

$$\sim 46 \text{ tablets}$$

Question 16- A 53-year-old female patient has been diagnosed with methicillin-resistant staphylococcus aureus (MRSA) bacteremia. The attending physician decides to infuse vancomycin 1250 mg IV over 1.5 hours. Its elimination rate constant is 0.12 hr^{-1} . The volume of distribution is 60 L.

Calculate the steady-state concentration.

Question 16- Solution: 115.7 mg/L

1. Calculate Infusion Rate (R_0):

$$R_0 = 1250 \text{ mg} / 1.5 \text{ hr} = 833.3 \text{ mg/hr}$$

2. Calculate Clearance (Cl):

$$k_e = \frac{Cl}{V_d} \rightarrow Cl = V_d \times k_e$$

$$Cl = 60 \text{ L} \times 0.12 \text{ hr}^{-1} = 7.2 \text{ L/hr}$$

3. Calculate Steady State Concentration (C_{ss}):

$$C_{ss} = \frac{F \times R_0}{Cl}$$

$$C_{ss} = \frac{833.3 \text{ mg/hr}}{7.2 \text{ L/hr}} = 115.7 \text{ mg/L}$$

Question 17- GH is a 72-year-old male patient (weight: 65 kg, SCr: 1.5 mg/dL) who has been prescribed vancomycin, which is to be given intravenously every 12 hours. The target steady-state average concentration is 10 mg/L. The drug is primarily renally cleared, and it is assumed that the drug clearance is equivalent to the creatinine clearance ($Cl \sim CrCl$).

Calculate the maintenance dose (D) for the patient, rounded to the nearest 50 mg.

Question 17- Solution: 300 mg

1. Calculate Creatinine Clearance (CrCl):

$$CrCl (\text{male}) = \frac{(140 - \text{age}) \times \text{Weight (kg)}}{72 \times \text{Serum Cr (mg/dL)}}$$

$$CrCl (\text{male}) = \frac{(140 - 72) \times 65 \text{ kg}}{72 \times 1.5 \text{ mg/dL}} = 41 \text{ mL/min}$$

2. Convert Clearance (Cl) to L/hr:

$$Cl = 40 \text{ mL/min} \times 0.06 = 2.46 \text{ L/hr}$$

3. Calculate Maintenance Dose (D):

Assuming $F = 1$ for intravenous administration:

$$C_{p_{av,ss}} = \frac{F \times D}{Cl \times \tau}$$

$$10 \text{ mg/L} = \frac{1 \times D}{2.4 \text{ L/hr} \times 12 \text{ hr}}$$

$$D = 10 \times 2.46 \times 12 = 295.2 \text{ mg}$$

4. Round to the nearest 50 mg:

$$295.2 \text{ mg} \rightarrow 300 \text{ mg}$$

Question 18- A 59-year-old female patient (weight: 180 lbs) is started on an intravenous infusion of heparin at 18 units/kg/hr. At your institution, heparin is available in bags of 25,000 units in 250 mL D5W.

Calculate the infusion rate in mL/hr and the duration (in hours) that one bag of heparin will last.

Question 18- Solution: 15 mL/hr, 17 hrs

1. Convert weight to kilograms:

$$\frac{180 \text{ lbs}}{2.2 \text{ lbs/kg}} = 81.8 \text{ kg}$$

2. Calculate Infusion Rate (Units):

Weight (kg) $\times$ Dose (units/kg/hr) = rate of infusion

$$81.8 \text{ kg} \times 18 \text{ units/kg/hr} = 1472.4 \text{ units/hr}$$

3. Calculate Infusion Rate (mL/hour):

$$\frac{1472 \text{ units}}{\text{hour}} \times \frac{250 \text{ mL}}{25000 \text{ units}} = 14.72 \text{ mL/hr}$$

4. Calculate Duration:

$$250 \text{ mL} / 14.7 \text{ mL/hr} = 17.006 \text{ hours}$$

Round to 17 hours

Question 19- You are preparing to dispense a pediatric suspension. The child weighs 26 lbs. The prescription is for azithromycin 200 mg/5 mL suspension, give 10 mg/kg on day 1, then 5 mg/kg on days 2–5. This concentration is available in a 15 mL, 22.5 mL or 30 mL bottle.

Calculate the total volume (mL) required for the 5-day course and determine which bottle size would be most appropriate.

Question 19- Solution: 8.87 mL, 15 mL bottle

1. Convert Weight to kg:

$$\text{Weight} = \frac{26 \text{ lbs}}{2.2 \text{ lbs/kg}} = 11.8 \text{ kg}$$

2. Calculate total amount (mg) required to fill Rx:

Day 1:

$$10 \frac{\text{mg}}{\text{kg}} \times 11.8 \text{ kg} = 118 \text{ mg}$$

Day 2 – 5:

$$5 \text{ mg/kg} \times 11.82 \text{ kg} = 59.1 \text{ mg/day} \times 4 \text{ days} = 236 \text{ mg}$$

$$\text{Total dose required: } 118 \text{ mg} + 236 \text{ mg} = 354 \text{ mg}$$

3. Simplify concentration of suspension:

$$\text{Concentration: } 200 \text{ mg/5 mL} = 40 \text{ mg/mL}$$

4. Calculate total volume required:

$$\frac{236 \text{ mg}}{40 \text{ mg/mL}} = 8.85 \text{ mL}$$

5. Select the most appropriate bottle to dispense:

15 mL

Question 20- A patient is currently prescribed drug A 100 mg once daily. Drug A has a half-life of 24 hours. The patient lost their medication and missed 5 doses, subsequently presenting to the Emergency Room with a drug level of 0 mg/L. To return the patient to steady state immediately, the clinician needs to provide a loading dose.

Calculate the loading dose of Drug A that is required (in mg).

Question 20- Solution: 200 mg

1. Calculate the elimination rate constant (k_e):

$$k_e = \frac{0.693}{t_{1/2}}$$

$$k_e = \frac{0.693}{24 \text{ hr}} = 0.0289 \text{ hr}^{-1}$$

2. Calculate the loading dose (D_L):

$$D_L = \frac{D_M}{1 - e^{-k\tau}}$$

$$D_L = \frac{100 \text{ mg}}{1 - e^{-0.0289 \text{ hr}^{-1} \times 24 \text{ hrs}}} = 200 \text{ mg}$$

Question 21- A 55-year-old male patient has been prescribed amiodarone IV 150 mg over 10 minutes, then 1 mg/min for 6 hours, then 0.5 mg/min for 18 hours. Supply: Bag 1 (for maintenance) is 450 mg in 250 mL of dextrose 5% in water (D5W).

Calculate the flow rate (mL/hr) for each phase of the infusion.

Question 21- Solution: Phase 1: 500 mL/hr. Phase 2: 33.3 mL/hr. Phase 3: 16.7 mL/hr.

1. Convert the maintenance dose rate to an hourly rate:

$$\text{Phase 1: } \frac{150 \text{ mg}}{10 \text{ min}} \times \frac{60 \text{ min}}{\text{hr}} = 900 \text{ mg/hr}$$

$$\text{Phase 2: } 1 \text{ mg/min} \times 60 \text{ min/hr} = 60 \text{ mg/hr}$$

$$\text{Phase 3: } 0.5 \text{ mg/min} \times 60 \text{ min/hr} = 30 \text{ mg/hr}$$

2. Calculate the concentration of the infusion:

$$\text{Concentration} = \frac{\text{Total Drug}}{\text{Total Volume}}$$

$$450 \text{ mg}/250 \text{ mL} = 1.8 \text{ mg/mL}$$

3. Calculate the flow rate:

$$\text{Phase 1: } \frac{900 \text{ mg}}{\text{hr}} \times \frac{\text{mL}}{1.8 \text{ mg}} = 500 \text{ mL/hr}$$

$$\text{Phase 2: } \frac{60 \text{ mg}}{\text{hr}} \times \frac{\text{mL}}{1.8 \text{ mg}} = 33.3 \text{ mL/hr}$$

$$\text{Phase 3: } \frac{30 \text{ mg}}{\text{hr}} \times \frac{\text{mL}}{1.8 \text{ mg}} = 16.7 \text{ mL/hr}$$

Question 22- MJ is a 45-year-old male who weighs 205 lbs. He is prescribed drug B 200 mg IV every 6 hours. Drug B has a volume of distribution of 0.5 L/kg and a clearance of 2.7 L/hr.

Calculate the steady-state peak concentration.

Question 22- Solution: 14.6 mg/L

1. Convert patient weight to kg:

$$\frac{205 \text{ lbs}}{2.2 \frac{\text{lbs}}{\text{kg}}} = 93.2 \text{ kg}$$

2. Calculate the volume of distribution:

$$V_d = 0.5 \frac{\text{L}}{\text{kg}} \times 93.2 \text{ kg} = 46.6 \text{ L}$$

3. Calculate k_e :

$$k_e = \frac{Cl}{V_d} = \frac{2.7 \text{ L/hr}}{46.6 \text{ L}} = 0.058 \text{ hr}^{-1}$$

4. Calculate the steady-state peak concentration:

$$C_{p_{\max,ss}} = \frac{F \times D}{V_d \times (1 - e^{-k \times \tau})}$$

$F = 1$, as drug is IV

$$C_{p_{\max,ss}} = \frac{1 \times 200 \text{ mg}}{46.6 \text{ L} \times (1 - e^{-0.058 \text{ hr}^{-1} \times 6 \text{ hrs}})} = 14.6 \text{ mg/L}$$

Question 23- A 23-year-old female patient is prescribed Carbolith® (lithium carbonate) 300 mg TID. Its bioavailability is 100%.

MW: Lithium (Li) = 7 g/mol. Carbon (C) = 12 g/mol. Oxygen (O) = 16 g/mol. Formula: Li_2CO_3 .

How many milliequivalents (mEq) of lithium does the patient receive daily?

Question 23- Solution: 24.3 mEq

1. Calculate total daily dose of lithium carbonate:

$$300 \text{ mg/tab} \times 3 \text{ tab/day} = 900 \text{ mg/day}$$

2. Calculate Molecular Weight (MW):

$$\text{Formula: } MW_{\text{Li}_2\text{CO}_3} = (2 \times MW_{\text{Li}}) + MW_{\text{C}} + (3 \times MW_{\text{O}})$$

$$\text{Calculation: } MW = (2 \times 7 \text{ g/mol}) + 12 \text{ g/mol} + (3 \times 16 \text{ g/mol}) = 74 \text{ g/mol}$$

3. Calculate mEq:

$$\text{mEq} = \frac{\text{mass of ion (mg)} \times \text{valence of ion}}{\text{molecular mass}}$$

4. Calculate the valence of Li_2CO_3 :

$$\text{Valence of Li} = 1$$

$$\times 2 \text{ lithium per lithium carbonate} = 2$$

5. Calculation:

$$\text{mEq} = \frac{900 \text{ mg} \times 2}{74 \text{ g/mol}} = 24.32 \text{ mEq}$$

Question 24- JD is a 62-year-old male patient who weighs 187 lbs. He has chronic asthma and has been prescribed theophylline for long-term symptom control. The patient is currently taking 400 mg of theophylline every 8 hours. Theophylline has a half-life of 7 hours and a volume of distribution of 0.5 L/kg. Its bioavailability is 80%.

Calculate the average steady-state plasma concentration in mg/L.

Question 24- Solution: 9.52 mg/L

1. Calculate k_e

$$k_e = \frac{0.693}{t_{1/2}} = \frac{0.693}{7 \text{ hrs}} = 0.099 \text{ hr}^{-1}$$

2. Convert weight to kg:

$$\frac{187 \text{ lbs}}{2.2 \text{ lbs/kg}} = 85 \text{ kg}$$

3. Calculate volume of distribution:

$$V_d = 0.5 \frac{\text{L}}{\text{kg}} \times 85 \text{ kg} = 42.5 \text{ L}$$

4. Calculate clearance:

$$Cl = k_e \times V_d = 0.099 \text{ hr}^{-1} \times 42.5 \text{ L} = 4.2 \text{ L/hr}$$

5. Calculate $C_{p,av,ss}$:

$$C_{p,av,ss} = \frac{F \times D}{Cl \times \tau}$$

$$C_{p,av,ss} = \frac{0.8 \times 400 \text{ mg}}{4.2 \frac{\text{L}}{\text{hr}} \times 8 \text{ hrs}} = 9.52 \text{ mg/L}$$

Question 25- A patient is prescribed amoxicillin oral solution 250 mg/5 mL for the treatment of a severe upper respiratory tract infection. The patient is instructed to take 1.5 tablespoons three times daily for 10 days.

Calculate the total daily dose of amoxicillin in milligrams.

Question 26- Solution: 3375 mg/day

1. Convert tablespoons to milliliters:

$$15 \text{ mL/tbsp} \times 1.5 \text{ tbsp} = 22.5 \text{ mL}$$

$$22.5 \frac{\text{mL}}{\text{dose}} \times 3 \frac{\text{doses}}{\text{day}} = 67.5 \text{ mL/day}$$

2. Simplify the concentration of the solution:

$$\frac{250 \text{ mg}}{5 \text{ mL}} = 50 \text{ mg/mL}$$

3. Calculate the total daily dose:

$$\frac{67.5 \text{ mL}}{\text{day}} \times \frac{50 \text{ mg}}{\text{mL}} = 3375 \text{ mg/day}$$

Question 26- A patient is currently taking Dilantin® (phenytoin) at a dose of 300 mg/day, achieving a steady-state plasma concentration of 8 mg/L. The clinician wishes to increase the dose to achieve a target concentration of 15 mg/L. The Michaelis-Menten constant is estimated to be 6 mg/L, and bioavailability is 100%.

Calculate the new daily dose (D) required to achieve the target steady-state concentration of 15 mg/L.

Question 26- Solution: 375 mg/day

1. Calculate Maximum Metabolism Speed ($V_{\max}$):

$$V_{\max} = \frac{(K_m + C_{p_{ss}}) \times F \times R_a}{C_{p_{ss}}}$$
$$V_{\max} = \frac{(6 \text{ mg/L} + 8 \text{ mg/L}) \times 1 \times 300 \text{ mg/day}}{8 \text{ mg/L}}$$
$$V_{\max} = \frac{14 \text{ mg/L} \times 300 \text{ mg/day}}{8 \text{ mg/L}} = 525 \text{ mg/day}$$

2. Calculate New Daily Dose (D):

Using target $C_{p_{ss}} = 15 \text{ mg/L}$, $V_{\max} = 525 \text{ mg/day}$, and $\tau = 1 \text{ day}$

$$\frac{F \times D}{\tau} = \frac{V_{\max} \times C_{p_{ss}}}{K_m + C_{p_{ss}}}$$
$$\frac{1 \times D}{1 \text{ day}} = \frac{525 \text{ mg/day} \times 15 \text{ mg/L}}{6 \text{ mg/L} + 15 \text{ mg/L}}$$
$$D = \frac{7875 \text{ mg/day} \cdot \text{mg/L}}{21 \text{ mg/L}} = 375 \text{ mg/day}$$

Question 27- MA is an 85-year-old female (weight: 50 kg, SCr: 1.6 mg/dL). She has been prescribed digoxin and her physician would like to target a plasma concentration of 0.8 ng/mL. The clearance is as follows: $Cl_{dig} = (0.33 \text{ mL/min/kg}) + (0.9 \times CrCl)$. Digoxin is available in 62.5 mcg and 125 mcg tablets and has a bioavailability of 70%.

Calculate the daily dose in mcg.

Question 27- Solution: 62.5 mcg

1. Calculate CrCl:

$$CrCl_{\text{female}} = \frac{(140 - \text{Age}) \times \text{Wt}}{72 \times \text{SCr}} \times 0.85$$

$$CrCl = \frac{(140 - 85) \times 50 \text{ kg}}{72 \times 1.6 \frac{\text{mg}}{\text{dL}}} \times 0.85 = 20.3 \text{ mL/min}$$

2. Calculate Digoxin Clearance:

$$Cl_{\text{dig}} = (0.33 \times \text{Wt}) + (0.9 \times CrCl)$$

$$Cl_{\text{dig}} = (0.33 \times 50 \text{ kg}) + (0.9 \times 20.3 \text{ mL/min}) = 34.8 \text{ mL/min}$$

3. Convert to L/day:

$$34.8 \text{ mL/min} \times \frac{1440 \text{ min}}{1 \text{ day}} \times \frac{1 \text{ L}}{1000 \text{ mL}} = 50.1 \text{ L/day}$$

4. Convert concentration to mcg/L:

$$1 \text{ ng/mL} = 1 \text{ mcg/L} \rightarrow 0.8 \text{ ng/mL} = 0.8 \text{ mcg/L}$$

5. Calculate Dose:

$$D = \frac{C_{ss} \times Cl}{F}$$

$$D = \frac{0.8 \text{ mcg/L} \times 50.1 \text{ L}}{0.7} = 57.2 \text{ mcg}$$

Round to nearest practical increment (62.5 mcg).

Question 28- GK is a 43-year-old female (height: 6 feet and 2 inches, weight: 210 lbs) who is interested in discussing pharmacotherapy options for weight loss. Prior to making any recommendations, you would like to gather more information.

Calculate her ideal body weight (IBW) and body mass index (BMI).

Question 28- Solution: 77.7 kg and 27.0 kg/m²

1. Convert Height to Inches:

$$12 \text{ inches/foot} \times 6 \text{ feet} = 72 \text{ inches}$$

$$6'2'' = 72 \text{ inches} + 2 \text{ inches} = 74 \text{ inches}$$

2. Calculate IBW:

$$\text{IBW}_{\text{female}} = 45.5 \text{ kg} + 2.3 \text{ kg} \times (74 - 60):$$

$$\text{IBW} = 45.5 \text{ kg} + (2.3 \times 14 \text{ inches}) = 45.5 \text{ kg} + 32.2 \text{ kg} = 77.7 \text{ kg}$$

3. Convert Height to Meters:

$$74 \text{ inches} \times 2.54 \text{ cm/inch} = 187.96 \text{ cm} = 188 \text{ cm} = 1.88 \text{ m}$$

4. Convert Weight to Kilograms:

$$210 \text{ lbs} / (2.2 \text{ lbs/kg}) = 95.45 \text{ kg}$$

5. Calculate BMI:

$$\text{BMI} = \frac{\text{Weight (kg)}}{(\text{Height (m)})^2}:$$

$$\text{BMI} = \frac{95.45 \text{ kg}}{(1.88 \text{ m})^2} \approx 27.0 \text{ kg/m}^2$$

Question 29- JT is a 55-year-old female patient who is being started on Dilantin® (phenytoin) for seizure prophylaxis. Given the drug's saturable metabolism and non-linear kinetics, the clinician wants to calculate the daily maintenance dose required to achieve a steady-state concentration of 15 mg/L. The Michaelis-Menten constant is 4 mg/L. The maximum rate of metabolism (V_{max}) is 500 mg/day. Bioavailability is 100%.

Calculate the required daily maintenance dose (D) in mg.

Question 29- Solution: 395 mg

1. Identify the formula for Phenytoin (Non-Linear Kinetics):

$$\frac{F \times D}{\tau} = \frac{V_{\max} \times C_{p_{ss}}}{K_m + C_{p_{ss}}}$$

2. Rearrange the equation to solve for Dose (D):

$$D = \frac{V_{\max} \times C_{p_{ss}} \times \tau}{(K_m + C_{p_{ss}}) \times F}$$

3. Calculate the maintenance dose required:

$$D = \frac{500 \text{ mg/day} \times 15 \text{ mg/L} \times 1 \text{ day}}{(4 \text{ mg/L} + 15 \text{ mg/L}) \times 1.0} = 395 \text{ mg}$$

Question 30- A 6-year-old female patient (weight: 44 lbs) is prescribed amoxicillin 90 mg/kg/day divided BID for the treatment of acute otitis media (AOM). You only have amoxicillin 250 mg/5 mL oral suspension in stock.

Calculate the volume (mL) to be administered per dose.

Question 30- Solution: 18 mL

1. Convert Weight to kg:

$$44 \text{ lbs} / (2.2 \frac{\text{lbs}}{\text{kg}}) = 20 \text{ kg}$$

2. Calculate Total Daily Dose:

$$90 \text{ mg/kg/day} \times 20 \text{ kg} = 1800 \text{ mg/day}$$

3. Divide for BID Dosing:

$$\frac{1800 \text{ mg/dose}}{2 \text{ doses/day}} = 900 \text{ mg per dose}$$

4. Simplify concentration of suspension:

$$\frac{250 \text{ mg}}{5 \text{ mL}} = 50 \frac{\text{mg}}{\text{mL}}$$

5. Calculate the volume per dose:

$$\frac{900 \text{ mg}}{50 \text{ mg/mL}} = 18 \text{ mL}$$

Question 31- A 62-year-old male patient is experiencing severe oral mucositis following a cycle of chemotherapy. The oncologist prescribes 16 fluid ounces of a "Magic Mouthwash" compounded oral rinse, which traditionally follows a 1:1:1 ratio of Xylocaine® (lidocaine) 2% viscous solution, Benadryl® (diphenhydramine) elixir, and Maalox® (magnesium hydroxide and aluminum hydroxide) suspension.

Calculate the final concentration (% w/v) of lidocaine in the compounded oral mouthwash.

Question 31- Solution: 0.67% w/v

1. Convert fluid ounces to mL:

$$29.57 \text{ mL/fluid ounce} \times 16 \text{ fluid ounces} = 473 \text{ mL}$$

2. Calculate Total Parts:

$$\text{Total Parts} = 1 \text{ (Xylocaine®)} + 1 \text{ (Benadryl®)} + 1 \text{ (Maalox®)} = 3 \text{ Parts}$$

3. Calculate Component Volume (V_i):

$$V_i = \frac{473 \text{ mL}}{3 \text{ parts}} = 157.67 \text{ mL/part} = 157.67 \text{ mL of lidocaine solution}$$

4. Calculate amount of lidocaine:

$$2\% \text{ (w/v) of } 157.67 \text{ mL} = 0.02 \times 157.67 \text{ mL} = 3.154 \text{ g}$$

5. Calculate final concentration:

$$\frac{3.154 \text{ g}}{473 \text{ mL}} = 0.0067 \frac{\text{g}}{\text{mL}} = 0.67 \frac{\text{g}}{100 \text{ mL}} = 0.67\% \left(\frac{\text{w}}{\text{v}} \right)$$

Question 32- A patient presents to your community pharmacy to refill their insulin pen. The prescription is for Lantus® (insulin glargine) 0.3 units/kg every evening at bedtime. The patient weighs 183 lbs. You will be supplying them with a box of 5 pens (3 mL each, 100 units/mL).

Calculate the days' supply, assuming an air shot of 2 units with every dose.

Question 32- Solution: 55 days

1. Convert Weight to kg:

$$183 \text{ lbs} / (2.2 \frac{\text{lbs}}{\text{kg}}) = 83.18 \text{ kg}$$

2. Calculate Daily Dose:

$$83.18 \text{ kg} \times 0.3 \text{ units/kg} = 24.95 \text{ units} \sim 25 \text{ units/dose}$$

$$25 \text{ units/dose} + 2 \text{ units/dose (priming)} = 27 \text{ units/dose}$$

3. Calculate the days' supply for each pen:

$$100 \text{ units/mL} \times 3 \text{ mL/pen} = 300 \text{ units/pen}$$

$$\frac{300 \text{ units/pen}}{27 \text{ units/dose}} = 11 \text{ doses/pen}$$

4. Calculate the days' supply for the box:

$$11 \frac{\text{doses}}{\text{pen}} \times 5 \text{ pens} = 55 \text{ doses}$$

$$1 \frac{\text{dose}}{\text{day}} \times 55 \text{ doses} = 55 \text{ days}$$

Question 33- MK is a 60-year-old female who has been admitted to the intensive care unit (ICU) for severe community-acquired pneumonia. She weighs 100 kilograms and is 5 feet and 5 inches tall. Her serum creatinine (SCr) is 120 $\mu\text{mol/L}$. Prior to choosing an antibiotic regimen, the attending physician would like to verify her renal function.

Calculate MK's creatinine clearance.

Question 33- Solution: 50.5 mL/min

1. Calculate ideal body weight (IBW):

$$\text{IBW} = 45.5 \text{ kg} + 2.3 \times (\# \text{ of inches over 5 feet})$$

$$\text{IBW} = 45.5 \text{ kg} + (2.3 \times 5 \text{ inches}) = 57 \text{ kg}$$

2. Decide which weight to use:

$\text{IBW} < \text{ABW}$, so we will not use ABW

$$130\% \text{ of IBW} = 1.3 \times \text{IBW} = 1.3 \times 57 \text{ kg} = 74.1 \text{ kg}$$

$\text{ABW} > 130\% \text{ of IBW}$, so we will use AdjBW

3. Calculate adjusted body weight (AdjBW):

$$\text{AdjBW} = \text{IBW} + 0.4 \times (\text{ABW} - \text{IBW})$$

$$\text{AdjBW} = 57 \text{ kg} + 0.4 \times (100 \text{ kg} - 57 \text{ kg})$$

$$\text{AdjBW} = 74.2 \text{ kg}$$

4. Calculate creatinine clearance (CrCl):

$$\text{CrCl (female)} = \frac{(140 - \text{age}) \times 1.2 \times \text{body weight (kg)}}{\text{serum creatinine } \left(\frac{\mu\text{mol}}{\text{L}}\right)} \times 0.85$$

$$\text{CrCl} = \frac{(140 - 60) \times 1.2 \times 74.2 \text{ kg}}{120 \mu\text{mol/L}} \times 0.85$$

$$\text{CrCl} = 50.5 \text{ mL/min}$$

Question 34- A pharmacy technician is preparing a compounded formulation using glycerin. To ensure the correct amount is added by weight, the technician needs to convert a measured volume into grams. The volume of glycerin used is 120 mL and its specific gravity (SG) is 1.25.

Calculate the weight of the glycerin, in grams.

Question 34- Solution: 150 g

1. Identify Weight of Water:

Based on the density of water (1g/mL), a volume of 120 mL of water weighs exactly 120 grams.

2. Rearrange for Weight:

$$SG = \frac{\text{Weight of substance}}{\text{Weight of equal volume of water}}$$

$$\text{Weight of substance} = SG \times \text{Weight of equal volume of water}$$

3. Calculate Final Weight:

$$\text{Weight} = 1.25 \times 120 \text{ g} = 150 \text{ g}$$

Question 35- A patient presents to your community pharmacy to refill their eye drops. The prescription is for Xalatan® (latanoprost) 1 drop in both eyes every evening at bedtime. Xalatan® is available in 2.5 mL bottles. There are 20 drops in every mL of Xalatan® solution.

Calculate the days' supply.

Question 35- Solution: 25 days

1. Calculate total number of drops:

$$2.5 \text{ mL} \times (20 \text{ drops/mL}) = 50 \text{ drops}$$

2. Daily Use:

$$\text{"1 gtt both eyes"} = 2 \text{ drops total per day}$$

3. Calculate days' supply:

$$50 \text{ drops} / (2 \text{ drops/day}) = 25 \text{ days}$$

Question 36- A patient is prescribed triamcinolone acetonide cream for a skin condition. They will be applying it to their entire back, which represents 18% of their total body surface area (BSA). Typically, application of this cream to 9% BSA requires 1 gram per application. The patient is instructed to apply the cream twice daily for 14 days. You have 30-gram tubes available in your pharmacy.

Calculate the total amount (in grams) and the number of tubes required to fill this prescription.

Question 36- Solution: 56 grams, 2 tubes

1. Determine grams per dose:

The back is approximately 18% BSA. Based on the dosing rule, $18\%/9\% = 2$. Therefore, 2 g of cream are needed per application.

2. Calculate total grams:

$$2 \text{ g/application} \times 2 \text{ applications/day} = 4 \text{ g/day}$$

$$4 \text{ g/day} \times 14 \text{ days} = 56 \text{ g}$$

3. Determine the number of tubes required:

The medication comes in 30 g tubes.

$$56 \text{ g}/(30 \text{ g/tube}) = 1.87 \text{ tubes} \sim 2 \text{ tubes}$$

Question 37- A 6-year-old pediatric patient (weight: 44 lbs) is admitted to the Intensive Care Unit (ICU) following a cardiovascular procedure. The physician orders a dopamine hydrochloride infusion to maintain hemodynamic stability. The pharmacy provides a premixed bag containing 400 mg of dopamine hydrochloride in 250 mL of 5% dextrose in water (D5W). The infusion pump is currently running at a rate of 4.5 mL/hour.

Determine the dosage the patient is receiving in mcg/kg/min.

Question 37- Solution: 6.0 mcg/kg/min.

1. Convert weight to kg:

$$44 \text{ lbs}/(2.2 \frac{\text{lbs}}{\text{kg}}) = 20 \text{ kg}$$

2. Calculate the concentration in mcg/mL:

$$\frac{400 \text{ mg}}{250 \text{ mL}} = 1.6 \text{ mg/mL}$$

$$1 \text{ mg} = 1000 \text{ mcg}$$

$$1.6 \text{ mg/mL} = 1600 \text{ mcg/mL}$$

3. Calculate the dosage based on rate and weight:

$$\frac{1600 \text{ mcg}}{\text{mL}} \times \frac{4.5 \text{ mL}}{\text{hr}} = 7200 \text{ mcg/hr}$$

$$\frac{7200 \text{ mcg}}{\text{hr}} \times \frac{1 \text{ hr}}{60 \text{ min}} = 120 \text{ mcg/min}$$

$$\frac{120 \text{ mcg/min}}{20 \text{ kg}} = 6 \text{ mcg/kg/min}$$

Question 38- A pharmacist receives a prescription to compound 60 g of a 2.5% hydrocortisone cream. The pharmacy inventory currently contains hydrocortisone 5% cream and hydrocortisone 1% cream.

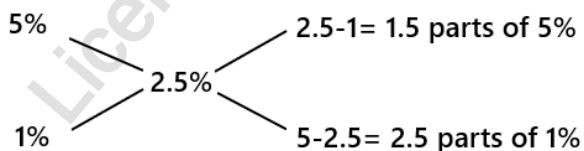
Determine the amount of each strength (in grams) required to prepare the mixture.

Question 38- Solution: 22.5 g of 5% cream and 37.5 g of 1% cream

1. Determine the number of parts for each concentration:

Subtract the lower concentration from the target to find parts of the higher concentration.

Subtract the target from the higher concentration to find parts of the lower concentration.



2. Calculate Total Parts:

$$\text{Total Parts} = 1.5 + 2.5 = 4.0 \text{ parts}$$

3. Calculate the quantity of 5% cream needed:

$$\text{Quantity}_{5\%} = \frac{1.5 \text{ parts}}{4.0 \text{ total parts}} \times 60 \text{ g} = 22.5 \text{ g}$$

4. Calculate the quantity of 1% cream needed:

$$\text{Quantity}_{1\%} = \frac{2.5 \text{ parts}}{4.0 \text{ total parts}} \times 60 \text{ g} = 37.5 \text{ g}$$

5. Verification:

$$22.5 \text{ g} + 37.5 \text{ g} = 60 \text{ g}$$

Question 39- JM is a 55-year-old male (height: 70 inches, weight: 100 kg, SCr: 1.2 mg/dL) admitted to the hospital for a severe gram-negative infection. The medical team plans to initiate therapy with gentamicin.

Calculate the patient's estimated Creatinine Clearance (CrCl) in mL/min.

Question 39- Solution: 82.4 mL/min

1. Calculate Ideal Body Weight (IBW):

$$\text{IBW (men)} = 50\text{kg} + 2.3 \text{ kg} \times (70 - 60 \text{ inches})$$

$$\text{IBW (men)} = 50 \text{ kg} + (2.3 \text{ kg} \times 10 \text{ inches}) = 73 \text{ kg}$$

2. Determine which weight to use:

Compare Actual Body Weight (100 kg) to IBW (73 kg).

Since $\text{IBW} < \text{ABW}$, we will not use ABW.

$$130\% \text{ of IBW} = 1.3 \times \text{IBW} = 1.3 \times 73 \text{ kg} = 94.9 \text{ kg}$$

Since $130\% \text{ of IBW} < \text{ABW}$, we must use AdjBW to calculate this patient's creatinine clearance (CrCl)

3. Calculate Adjusted Body Weight:

$$\text{AdjBW} = \text{IBW} + 0.4 \times (\text{Actual Body Weight} - \text{IBW})$$

$$\text{AdjBW} = 73 \text{ kg} + (0.4 \times (100 \text{ kg} - 73 \text{ kg}))$$

$$\text{AdjBW} = 73 \text{ kg} + (0.4 \times 27 \text{ kg})$$

$$\text{AdjBW} = 73 \text{ kg} + 10.8 \text{ kg} = 83.8 \text{ kg}$$

4. Calculate Creatinine Clearance:

$$\text{CrCl (mL/min)} = \frac{(140 - \text{Age}) \times \text{Weight (kg)}}{72 \times \text{SCr (mg/dL)}}$$

$$\text{CrCl} = \frac{(140 - 55) \times 83.8 \text{ kg}}{72 \times 1.2 \text{ mg/dL}}$$

$$\text{CrCl} = \frac{85 \times 83.8}{86.4}$$

$$\text{CrCl} = 82.4 \text{ mL/min}$$

Question 40- A 55-year-old palliative care patient is currently receiving high-dose oral opioid therapy using MS Contin® (morphine) controlled-release tablets. He takes 60 mg PO Q12H. Due to developing dysphagia, the clinical team decides to switch the patient to parenteral Dilaudid® (hydromorphone).

Calculate the maximum appropriate total daily dose of Dilaudid® (hydromorphone) IV in mg/day.

Question 40- Solution: 4.5 mg/day

1. Find reference data:

Morphine Oral Equivalence: 30 mg

Hydromorphone Parenteral Equivalence: 1.5 mg

2. Determine Total Daily Dose (TDD) of oral morphine:

$$60 \text{ mg/dose} \times 2 \text{ doses/day} = 120 \text{ mg/day}$$

3. Calculate Equianalgesic Hydromorphone IV Dose:

Using the ratio $\frac{\text{Target}_{\text{IV}}}{\text{Current}_{\text{PO}}}$:

$$\text{Dose}_{\text{new}} = \text{Dose}_{\text{current}} \times \frac{1.5 \text{ mg Hydromorphone IV}}{30 \text{ mg Morphine PO}}$$

$$\text{Dose}_{\text{new}} = 120 \text{ mg} \times \frac{1.5}{30}$$

$$\text{Dose}_{\text{new}} = 120 \times 0.05 = 6 \text{ mg/day}$$

4. Apply Cross-Tolerance Reduction:

Reduction: 25% since question specifies 'maximum' appropriate TDD

Calculation: $6 \text{ mg} \times (1 - 0.25) = 4.5 \text{ mg}$ of IV hydromorphone per day

Question 41- A pharmacist is compounding a topical anti-inflammatory ointment using 2.5 grams of Hydrocort® (hydrocortisone) powder and 122.5 grams of Aquaphor® (petrolatum) ointment base. The pharmacist combines the ingredients to create a homogeneous mixture.

Calculate the final concentration of hydrocortisone in the compounded preparation as a percentage weight per weight (% w/w).

Question 41- Solution: 2% w/w

1. Determine the Total Weight of the Preparation:

The total weight is the sum of the active ingredient (solute) and the base (solvent/diluent).

Total Weight = Weight of hydrocortisone + Weight of ointment base

$$\text{Total Weight} = 2.5 \text{ g} + 122.5 \text{ g} = 125 \text{ g}$$

2. Calculate the % w/w:

$$\%w/w = \frac{\text{g solute}}{100 \text{ g preparation}}$$

$$\% w/w = \frac{2.5 \text{ g}}{125 \text{ g}} \times 100\%$$

$$\% w/w = 0.02 \times 100\%$$

$$\% w/w = 2\%$$

Question 42- JP is a 54-year-old male patient admitted to the hospital with signs of hypokalemia. The attending physician prescribes 20 milliequivalents of potassium chloride to be given immediately to replenish electrolyte levels. The atomic weight of potassium (K) is 39 g/mol. The atomic weight of chloride (Cl) is 35.5 g/mol.

Calculate the equivalent mass of potassium chloride in milligrams (mg).

Question 42- Solution: 1490 mg

1. Determine Molecular Weight (MW):

$$\text{MW} = 39 + 35.5 = 74.5 \text{ g/mol}$$

2. Identify Valence:

Potassium chloride dissociates into K^+ and Cl^- . The valence is 1.

3. Rearrange Formula and Calculate:

$$\text{mEq} = \frac{\text{Mass (mg)} \times \text{Valence}}{\text{MW}}$$

$$\text{Mass (mg)} = \frac{\text{mEq} \times \text{MW}}{\text{Valence}}$$

$$\text{Mass (mg)} = \frac{20 \text{ mEq} \times 74.5 \text{ g/mol}}{1}$$

$$\text{Mass (mg)} = 1490 \text{ mg}$$

Question 43- JM is a 55-year-old patient taking Aspirin® (acetylsalicylic acid) for secondary prevention of myocardial infarction. A researcher is analyzing the chemical properties of the drug to determine its absorption characteristics in the stomach. Acid dissociation constant (K_a): 3.0×10^{-4} .

Calculate the pK_a of the drug.

Question 43- Solution: 3.52

1. Apply the pK_a definition:

$$\text{pK}_a = -\log K_a$$

$$\text{pK}_a = -\log(3.0 \times 10^{-4})$$

2. Expand using log properties:

$$\text{pKa} = -(\log 3.0 + \log 10^{-4})$$

3. Calculate:

$$\text{pKa} = -(0.477 + (-4))$$

$$\text{pKa} = -(-3.523)$$

$$\text{pKa} \approx 3.52$$

Question 44- A 45-year-old patient with short bowel syndrome is starting on total parenteral nutrition (TPN). The dietician has ordered a specific volume of dextrose to meet the patient's daily caloric requirements. The TPN order is as follows: 600 mL of dextrose 50%, 1000 mL of Travasol® (amino acids) and 250 mL of Intralipid® (fat emulsion).

Calculate the total kilocalories (kcal) provided by the dextrose component in this admixture.

Question 44- Solution: 1020 kcal

1. Establish values:

$$\text{Carbohydrates (Dextrose)} = 3.4 \text{ kcal/gram}$$

2. Determine the mass of dextrose:

Calculate the grams of dextrose present in 600 mL of a 50% solution.

$$\% \text{ w/v} = \frac{\text{grams of solute}}{100 \text{ mL of solution}}$$

$$50\% \text{ w/v} \rightarrow \frac{50 \text{ g}}{100 \text{ mL}}$$

$$\text{Total grams} = 600 \text{ mL} \times \frac{50 \text{ g}}{100 \text{ mL}} = 300 \text{ grams}$$

3. Calculate caloric content:

Multiply the grams of dextrose by the caloric value of carbohydrates in TPN (3.4 kcal/g).

$$\text{Total kcal} = 300 \text{ g} \times 3.4 \text{ kcal/g} = 1020 \text{ kcal}$$

Question 45- JK is a 50-year-old male patient (weight: 80 kg, height: 180 cm) admitted for a scheduled cardiovascular procedure. He has a past medical history of dyslipidemia which is currently managed with Lipitor® (atorvastatin). The clinical pharmacist is consulted to estimate the patient's nutritional requirements.

Calculate the patient's Basal Energy Expenditure (BEE) in kcal/day using the Harris-Benedict equation.

Question 45- Solution: 1729 kcal/day

1. Substitute values into the equation:

$$\text{BEE} = 66.67 + (13.75 \times \text{kg}) + (5 \times \text{cm}) - (6.76 \times \text{age})$$

$$\text{BEE} = 66.67 + (13.75 \times 80) + (5 \times 180) - (6.76 \times 50)$$

2. Solve for components:

$$\text{BEE} = 66.67 + (1100) + (900) - (338)$$

$$\text{BEE} = 2066.67 - 338$$

$$\text{BEE} = 1728.67$$

3. Round to nearest whole number:

$$\text{BEE} \approx 1729 \text{ kcal/day}$$

Question 46- JL is a 55-year-old male patient recovering from elective surgery. He is currently confined to his bed. He takes Centrum® (multivitamin) to support his nutritional status. A dietitian has already performed an assessment and calculated his Basal Energy Expenditure (BEE) to be 1600 kcal/day.

Calculate the patient's Total Daily Expenditure (TDE) in kcal/day.

Question 46- Solution: 2304 kcal/day

1. Determine Factors: Using the standard Total Daily Expenditure reference table:

Activity Factor (Confined to bed): 1.2

Stress Factor (Surgery): 1.2

2. Calculate: Apply the factors to the provided BEE.

$$\text{TDE} = \text{BEE} \times \text{Activity Factor} \times \text{Stress Factor}$$

$$\text{TDE} = 1600 \text{ kcal/day} \times 1.2 \times 1.2$$

$$\text{TDE} = 1600 \times 1.44$$

$$\text{TDE} = 2304 \text{ kcal/day}$$

Question 47- TK is a 4-year-old patient brought to the clinic for the treatment of a bacterial infection. The physician intends to prescribe ceftazidime and requests a dose calculation. The standard adult dose is 500 mg.

Suppose that a pediatric reference is currently unavailable. Calculate the appropriate pediatric dose for this patient, in mg.

Question 47- Solution: 125 mg

1. Apply Young's Rule:

$$\text{Pediatric dose} = \text{Average adult dose} \times \frac{\text{age (years)}}{\text{age (years)} + 12}$$

$$\text{Pediatric dose} = 500 \text{ mg} \times \frac{4 \text{ years}}{4 \text{ years} + 12 \text{ years}} = 500 \text{ mg} \times \frac{4 \text{ years}}{16 \text{ years}} = 125 \text{ mg}$$

Question 48- JR is a 5-year-old male patient (weight: 45 lbs) who requires antibiotic treatment for community-acquired pneumonia. The physician has prescribed cefepime and the standard adult dose for this condition is 500 mg.

Calculate the pediatric dose for JR, using Clark's Rule.

Question 48- Solution: 150 mg

1. Calculate:

$$\text{Pediatric dose} = \text{dose for average adult} \times \frac{\text{mass of child (lbs)}}{150}$$

$$\text{Pediatric dose} = 500 \text{ mg} \times \frac{45 \text{ lbs}}{150} = 150 \text{ mg}$$

Question 49- JL is a 55-year-old patient receiving treatment for osteoarthritis with Aleve® (naproxen). The pharmacist wishes to evaluate the drug's absorption potential in the stomach. Naproxen is a weak acid. The pKa of naproxen is 4.2. Gastric pH is 2.2.

Calculate the percentage of the drug that is ionized in the gastric environment.

Question 49- Solution: 0.99%

1. Substitute values into the Weak Acid equation:

$$\text{Weak acid \% ionization} = \frac{100}{1 + 10^{(\text{pKa} - \text{pH})}}$$

$$\text{Weak acid \% ionization} = \frac{100}{1 + 10^{(4.2 - 2.2)}}$$

2. Calculate the exponent term:

$$\text{pKa} - \text{pH} = 4.2 - 2.2 = 2$$

$$10^2 = 100$$

3. Final Calculation:

$$\% \text{ Ionized} = \frac{100}{1 + 100} = \frac{100}{101} \approx 0.99\%$$

Note: Since the pH is lower than the pKa (acidic environment), a weak acid will exist primarily in its non-ionized form, consistent with the low percentage calculated.

Question 50- A researcher is analyzing the pharmacokinetics of Xylocaine® (lidocaine) using the well-stirred model of hepatic elimination. The study aims to predict the hepatic clearance of the drug based on liver physiology and drug characteristics. The hepatic blood flow (Q) is 1500 mL/min. The intrinsic clearance is 300 L/hr. The fraction unbound is 0.3.

Calculate the estimated hepatic clearance in L/min.

Question 50- Solution: 0.75 L/min

1. Convert hepatic blood flow (Q) to L/min:

$$1500 \frac{\text{mL}}{\text{min}} \times \frac{1 \text{ L}}{1000 \text{ mL}} = 1.5 \text{ L/min}$$

2. Convert intrinsic clearance to L/min:

$$300 \frac{\text{L}}{\text{hr}} \times \frac{1 \text{ hr}}{60 \text{ min}} = 5 \text{ L/min}$$

3. Substitute values into the Well-Stirred equation:

$$\text{Cl}_H = \frac{Q \times f_u \times \text{Cl}_{\text{int}}}{Q + f_u \times \text{Cl}_{\text{int}}}$$

$$\text{Cl}_H = \frac{1.5 \frac{\text{L}}{\text{min}} \times 0.3 \times 5.0 \text{ L/min}}{1.5 \text{ L/min} + (0.3 \times 5.0 \frac{\text{L}}{\text{min}})} = 0.75 \text{ L/min}$$

Question 51- A compounding pharmacist is preparing a capsule formulation containing atropine sulfate. The pharmacy uses a balance to weigh the ingredients. To ensure pharmaceutical accuracy, the pharmacist must determine the smallest amount that can be weighed without exceeding the maximum permissible error. The sensitivity requirement (SR) is 6 mg and the maximum acceptable error is 5%.

Determine the minimum weighable quantity (MWQ) for this balance, in mg.

Question 51- Solution: 120 mg

1. Apply Formula:

$$\text{MWQ} = \frac{\text{Sensitivity}}{\text{Error (0.05)}}$$

$$\text{MWQ} = \frac{6 \text{ mg}}{0.05} = 120 \text{ mg}$$

Question 52- An anesthesiologist is administering Marcaine® (bupivacaine) to a patient undergoing a minor surgical procedure. Bupivacaine is a weak base with a pKa of 8.1. The tissue at the site of injection has a measured pH of 7.1 due to local inflammation.

Calculate the percent ionization of bupivacaine in this tissue environment.

Question 52- Solution: 90.9%

1. Substitute values into the equation:

$$\% \text{ ionization (weak base)} = \frac{100}{1 + 10^{(\text{pH} - \text{pKa})}}$$

$$\% \text{ ionization (weak base)} = \frac{100}{1 + 10^{(7.1 - 8.1)}}$$

2. Final Calculation:

$$\% \text{ Ionized} = \frac{100}{1.1} = 90.9\%$$

Question 53- PT is a 32-year-old male patient with Type 1 diabetes mellitus who has been prescribed Humalog KwikPen® (insulin lispro) for mealtime glycemic control. PT is instructed to inject 10 units subcutaneously three times daily with meals. He is also instructed to dial up and expel an "air shot" of 2 units before every injection to ensure the needle is patent and free of air bubbles. The pharmacy dispenses a box of pens; each pen contains 3 mL of insulin at a concentration of 100 units/mL.

Calculate the number of full days a single pen will last given this specific administration technique.

Question 53- Solution: 8 days

1. Calculate Total Units per Pen:

$$3 \text{ mL} \times 100 \text{ units/mL} = 300 \text{ units}$$

1. Calculate Units Used Per Administration Event:

Include both the therapeutic dose and the loss from priming (air shot).

$$\text{Units Per Event} = \text{Dose} + \text{Prime}$$

$$U_{\text{event}} = 10 \text{ units (dose)} + 2 \text{ units (prime)} = 12 \text{ units}$$

3. Calculate Total Daily Usage:

$$\text{Total Daily Usage} = \text{Units Per Event} \times \text{Frequency}$$

$$U_{\text{daily}} = 12 \text{ units/dose} \times 3 \text{ doses/day} = 36 \text{ units/day}$$

4. Calculate Duration:

$$\text{Days} = \frac{\text{Total Units in Pen}}{\text{Total Daily Usage}}$$

$$\text{Days} = \frac{300 \text{ units}}{36 \text{ units/day}} = 8.33 \text{ days}$$

5. Determine Full Days: Since the patient requires 36 units for a complete day of therapy, the pen will be insufficient during the 9th day.

Answer = 8 full days

Question 54- A pharmacist is performing quality assurance on a compounded batch of Robitussin® (guaifenesin) syrup. To verify the concentration, the pharmacist weighs 120 mL of the liquid. She obtains a result of 150 g.

Calculate the specific gravity (SG) of the syrup.

Question 54- Solution: 1.25

1. Determine the weight of an equal volume of water:

Recall that the density of water is 1 g/mL.

Therefore, 120 mL of water weighs 120 g.

2. Calculate:

$$\text{SG} = \frac{\text{Weight of substance}}{\text{Weight of equal volume of water}}$$

$$\text{SG} = \frac{150 \text{ g}}{120 \text{ g}} = 1.25$$

Note: Specific gravity is a ratio and therefore has no units.

Question 55- AR is a 34-year-old male patient presenting to the pharmacy with a prescription for prednisone. The prescription is written for 5 mg tablets with a specific tapering schedule to prevent adrenal insufficiency. The prescribed schedule is as follows: 40 mg PO daily for 3 days, 30 mg PO daily for 3 days, 20 mg PO daily for 3 days, 10 mg PO daily for 3 days, 5 mg PO daily for 3 days then discontinue.

Calculate the total number of 5 mg tablets required to fulfill this prescription.

Question 55- Solution: 63 tablets

1. Calculate tablets for the 40 mg tier:

$$\frac{40 \text{ mg/day}}{5 \text{ mg/tab}} \times 3 \text{ days} = 8 \text{ tabs/day} \times 3 \text{ days} = 24 \text{ tablets}$$

2. Calculate tablets for the 30 mg tier:

$$\frac{30 \text{ mg/day}}{5 \text{ mg/tab}} \times 3 \text{ days} = 6 \text{ tabs/day} \times 3 \text{ days} = 18 \text{ tablets}$$

3. Calculate tablets for the 20 mg tier:

$$\frac{20 \text{ mg/day}}{5 \text{ mg/tab}} \times 3 \text{ days} = 4 \text{ tabs/day} \times 3 \text{ days} = 12 \text{ tablets}$$

4. Calculate tablets for the 10 mg tier:

$$\frac{10 \text{ mg/day}}{5 \text{ mg/tab}} \times 3 \text{ days} = 2 \text{ tabs/day} \times 3 \text{ days} = 6 \text{ tablets}$$

5. Calculate tablets for the 5 mg tier:

$$\frac{5 \text{ mg/day}}{5 \text{ mg/tab}} \times 3 \text{ days} = 1 \text{ tab/day} \times 3 \text{ days} = 3 \text{ tablets}$$

6. Sum the total quantity:

$$24 + 18 + 12 + 6 + 3 = 63 \text{ tablets}$$

Question 56- A dermatologist prescribes Diprolene® (betamethasone dipropionate) ointment for a patient with psoriasis. The specific dosing instruction requires the application of a thin layer, defined as 2 mg/cm^2 , to the affected areas twice daily. The patient's body surface area (BSA) is 1.8 m^2 and 5% of their BSA is affected by psoriasis. You will be dispensing a 60 g tube of Diprolene®.

Calculate the days' supply for this prescription.

Question 56- Solution: 16 days

1. Calculate Affected Area in m^2 :

$$\text{Area}_{\text{affected}} = 1.8 \text{ m}^2 \times 0.05 = 0.09 \text{ m}^2$$

2. Convert Area to cm²:

$$0.09 \text{ m}^2 \times \frac{10,000 \text{ cm}^2}{1 \text{ m}^2} = 900 \text{ cm}^2$$

3. Calculate Dose per Application:

Using the density rule of 2 mg/cm²:

$$\text{Dose}_{\text{app}} = 900 \text{ cm}^2 \times 2 \text{ mg/cm}^2 = 1800 \text{ mg}$$

Convert to grams (1000 mg = 1 g):

$$1800 \text{ mg} = 1.8 \text{ g}$$

4. Calculate Daily Usage:

$$1.8 \text{ g/dose} \times 2 \text{ doses/day} = 3.6 \text{ g/day}$$

5. Calculate Days Supply:

$$\frac{60 \text{ g}}{3.6 \text{ g/day}} = 16.67 \text{ days}$$

Rounding down to the nearest full day of complete therapy: 16 days

Question 57- JR is a 45-year-old male who is currently stabilized on Dilantin® (phenytoin sodium) capsules at a dose of 300 mg per day for seizure management. Due to recent difficulty swallowing, his physician wishes to switch him to Dilantin® (phenytoin) oral suspension. The salt factor (S) of phenytoin sodium capsules is 0.92 and the salt factor (S) of phenytoin oral suspension is 1.0.

Calculate the exact mathematical equivalent daily dose of the oral suspension required to maintain the same amount of active drug.

Question 57- Solution: 276 mg

1. Calculate the active drug from the current capsule dose:

Because the salt factor is 0.92, the capsule is 92% active drug.

$$\text{Active Drug} = \text{Dose} \times S$$

$$\text{Active Drug} = 300 \text{ mg} \times 0.92 = 276 \text{ mg}$$

2. Calculate the required dose for the new liquid formulation:

The oral suspension is formulated as phenytoin acid, which has a salt factor of 1.0 (meaning it is 100% active drug).

$$\text{Dose}_{\text{new}} \times S_{\text{new}} = \text{Dose}_{\text{old}} \times S_{\text{old}}$$

$$\text{Dose}_{\text{new}} = \frac{276 \text{ mg}}{1.0} = 276 \text{ mg}$$

3. Conclusion:

To provide the exact same amount of active phenytoin base, the patient requires 276 mg of the oral suspension daily.

Question 58- JW is a 55-year-old male patient (weight: 165 lbs) with type 2 diabetes mellitus who is currently taking Lantus® (insulin glargine) 0.2 units/kg subcutaneously at bedtime. Despite an adequate fasting blood glucose, his postprandial glucose levels remain elevated. The clinician decides to switch JW to a standard basal-bolus insulin regimen using Lantus® (insulin glargine) and Humalog® (insulin lispro) while maintaining the same Total Daily Dose (TDD). The new regimen will consist of 50% basal insulin and 50% short-acting bolus insulin divided equally across three daily meals.

Calculate the required dose of Humalog® (insulin lispro) to be administered before each meal. Do not account for air shots.

Question 58- Solution: 2.5 units

1. Convert weight to kg:

$$\frac{165 \text{ lbs}}{2.2 \text{ lbs/kg}} = 75 \text{ kg}$$

2. Calculate the total daily dose (TDD):

$$0.2 \frac{\text{units}}{\text{kg}} \times 75 \text{ kg} = 15 \text{ units}$$

3. Calculate Basal and Bolus Requirements:

The standard basal-bolus distribution is 50% basal and 50% bolus.

Basal requirement = $0.50 \times 15 \text{ units} = 7.5 \text{ units}$ of Lantus® (insulin glargine) daily.

Total daily bolus requirement = $0.50 \times 15 = 7.5$ units of Humalog® (insulin lispro) daily.

4. Calculate Short-Acting Mealtime Dose:

The total daily bolus dose is divided equally among 3 main meals.

$$\text{Mealtime Dose} = \frac{7.5 \text{ units/meal}}{3 \text{ meals}} = 2.5 \text{ units/meal}$$

Question 59- A pharmacist receives a prescription for Tobradex® (tobramycin/dexamethasone) ophthalmic suspension. The prescription instructions read: "instill 2 drops in each eye QID for 7 days." The calibrated dropper provided with the bottle dispenses 20 drops per milliliter.

Calculate the total volume of medication required to fill this prescription.

Question 59- Solution: 5.6 mL

1. Determine daily drop requirement:

$$\text{Daily drops} = 2 \text{ drops} \times 2 \text{ eyes} \times 4 \text{ times/day} = 16 \text{ drops/day}$$

2. Calculate total drops for duration:

$$16 \text{ drops/day} \times 7 \text{ days} = 112 \text{ total drops}$$

3. Convert drops to milliliters:

$$\text{Volume} = \frac{112 \text{ drops}}{20 \text{ drops/mL}}$$

$$\text{Volume} = 5.6 \text{ mL}$$

Question 60- A 62-year-old patient with a ventricular arrhythmia is administered a single IV bolus dose of Xylocaine® (lidocaine). The pharmacokinetics of this drug are best described by a two-compartment open model. The distribution coefficient (C_1) is 10 mg/L, the distribution rate constant (α) is 2.0 h^{-1} , the elimination coefficient (C_2) is 2.0 mg/L and the elimination rate constant (β) is 0.1 h^{-1} .

Calculate the plasma concentration (C) of the drug at 1.5 hours after administration.

Question 60- Solution: 2.22 mg/L

1. Substitute values into the two-compartment equation:

$$C = C_1 \times e^{-\alpha \times t} + C_2 \times e^{-\beta \times t}$$

$$C = 10 \times e^{-(2.0 \times 1.5)} + 2.0 \times e^{-(0.1 \times 1.5)}$$

2. Calculate the exponents:

$$C = 10 \times e^{-3} + 2.0 \times e^{-0.15}$$

3. Solve for individual terms:

$$C = 10 \times (0.0498) + 2.0 \times (0.8607)$$

$$C = 0.498 + 1.7214 = 2.22 \text{ mg/L}$$

Question 61- A compounding pharmacist needs to prepare an acetate buffer solution to maintain stability for a liquid formulation. The solution requires mixing acetic acid (weak acid) and sodium acetate (conjugate base). The desired pH is 5.06. The pKa of acetic acid is 4.76. The concentration of acetic acid is 0.2 M.

Calculate the required concentration of sodium acetate in M.

Question 61- Solution: 0.4 M

1. Substitute values into the Henderson-Hasselbalch equation:

$$\text{pH} = \text{pKa} + \log\left(\frac{[\text{A}^-]}{[\text{HA}]}\right)$$

$$5.06 = 4.76 + \log\left(\frac{[\text{A}^-]}{0.2}\right)$$

2. Rearrange to isolate the log term:

$$\log\left(\frac{[\text{A}^-]}{0.2}\right) = 5.06 - 4.76$$

$$\log\left(\frac{[\text{A}^-]}{0.2}\right) = 0.3$$

3. Solve for the concentration ratio using antilog:

$$\frac{[A^-]}{0.2} = 10^{0.3}$$

$$\frac{[A^-]}{0.2} \approx 2$$

4. Calculate the final concentration:

$$[A^-] = 2 \times 0.2 = 0.4 \text{ M}$$

Question 62- A patient admitted for an intra-abdominal infection has been prescribed an intravenous infusion of 100 mL of Flagyl® (metronidazole), infused over 30 minutes. The nurse is setting up the gravity infusion set and needs to determine the correct drip rate to ensure the medication is administered over the prescribed duration. The IV set drop factor is 15 drops/mL.

Calculate the required infusion flow rate in drops/min.

Question 62- Solution: 50 drops/min

1. Calculate:

$$\text{Flow Rate (drops/min)} = \frac{\text{Volume (mL)} \times \text{Drop Factor (drops/mL)}}{\text{Time (min)}}$$

$$\text{Flow Rate} = \frac{100 \text{ mL} \times 15 \text{ drops/mL}}{30 \text{ min}} = 50 \text{ drops/min}$$

Question 63- A hospital pharmacist receives an order to prepare 500 mL of a 0.5% w/v chlorhexidine gluconate irrigation solution for a surgical procedure. The pharmacy currently has chlorhexidine gluconate 4% w/v stock solution and sterile water for irrigation available.

Calculate the volume of the stock solution required to prepare this prescription, in mL.

Question 63- Solution: 62.5 mL

1. Set up the equation:

$$C_i V_i = C_f V_f$$

$$4\% \times V_i = 0.5\% \times 500 \text{ mL}$$

2. Solve for V_i :

$$V_i = \frac{0.5 \times 500}{4} = 62.5 \text{ mL}$$

Question 64- A 72-year-old male patient with atrial fibrillation is receiving anticoagulation therapy with Coumadin® (warfarin) for stroke prevention. The patient currently takes a maintenance dose of 5 mg daily. Upon review of the latest laboratory results, the clinician notes that the patient's INR is 1.6, which is subtherapeutic (target INR: 2-3). They decide to adjust the dose according to the clinic's protocol, which is to increase the total weekly dose by 15%.

Calculate the daily dose required (note: the daily dose may change throughout the week).

Question 64- Solution: 6 mg Monday-Friday, 5 mg Saturday & Sunday

1. Calculate the current weekly dose:

$$5 \text{ mg/day} \times 7 \text{ days} = 35 \text{ mg/week}$$

2. Calculate the new weekly dose:

$$35 \text{ mg} \times (1 + 0.15) = 35 \text{ mg} \times 1.15$$

$$35 \times 1.15 = 40.25 \text{ mg} \sim 40 \text{ mg}$$

3. Determine the daily dose required to achieve this weekly dose:

$$5 \text{ mg/day} \times 7 \text{ days} = 35 \text{ mg/week} \rightarrow \text{dose too low}$$

$$6 \text{ mg/day} \times 7 \text{ days} = 42 \text{ mg/week} \rightarrow \text{dose too high}$$

$$6 \text{ mg/day} \times 5 \text{ days} = 30 \text{ mg}$$

$$5 \text{ mg/day} \times 2 \text{ days} = 10 \text{ mg}$$

$$30 \text{ mg} + 10 \text{ mg} = 40 \text{ mg}$$

Can do 6 mg per day Monday-Friday, then 5 mg per day Saturday & Sunday

Question 65- A clinical researcher is evaluating the bioavailability of a new oral suspension of amoxicillin. During the absorption phase, the researcher determines the amount of drug remaining in the gastrointestinal tract and the specific rate at which the drug is entering the systemic circulation. There is 400 mg of amoxicillin in the GI tract, and the rate of absorption is 320 mg/hr.

Calculate the first-order absorption rate constant in hr^{-1} .

Question 65- Solution: 0.8 hr^{-1}

1. Identify Known Variables:

$$\text{Rate of absorption } \left(\frac{dAb}{dt}\right) = 320 \text{ mg/hr}$$

$$\text{Amount in GI tract } (A_{GI}) = 400 \text{ mg}$$

2. Rearrange the Equation:

$$\frac{dAb}{dt} = A_{GI} \times k_a$$

$$k_a = \frac{\frac{dAb}{dt}}{A_{GI}}$$

3. Calculate:

$$k_a = \frac{320 \text{ mg/hr}}{400 \text{ mg}} = 0.8 \text{ hr}^{-1}$$

Question 66- JL is a 55-year-old male admitted for a severe gram-negative infection. The clinical pharmacist is designing a dosing regimen for gentamicin. The goal is to achieve specific steady-state peak and trough concentrations to ensure efficacy while minimizing toxicity. The target steady-state peak is 8 mg/L, the target steady-state trough is 1 mg/L, and the elimination half-life is 3 hours.

Determine the dosing interval required to maintain these steady-state parameters, in hours.

Question 66- Solution: 9 hours

1. Calculate the elimination rate constant (k):

$$k = \frac{0.693}{t_{1/2}}$$

$$k = \frac{0.693}{3 \text{ h}} = 0.231 \text{ h}^{-1}$$

2. Set up the fluctuation equation to solve for τ :

$$\text{Fluctuation: } \frac{C_{p_{\max,ss}}}{C_{p_{\min,ss}}} = e^{k \times \tau}$$

$$\frac{8 \text{ mg/L}}{1 \text{ mg/L}} = e^{0.231 \times \tau}$$

$$8 = e^{0.231 \times \tau}$$

3. Take the natural log (ln) of both sides:

$$\ln(8) = \ln(e^{0.231 \times \tau})$$

$$2.079 = 0.231 \times \tau$$

4. Solve for τ :

$$\tau = \frac{2.079}{0.231} \approx 9 \text{ hours}$$

Question 67- A 78-year-old female patient is admitted to the hospital with a complicated urinary tract infection (UTI). The clinical team decides to initiate a continuous intravenous infusion of tobramycin. The pharmacokinetic profile of the patient indicates a specific elimination half-life of 6 hours.

Calculate the time (in hours) required for the plasma concentration to reach 95% of the steady-state concentration.

Question 67- Solution: 25.9 hours

1. Calculate the elimination rate constant (k):

$$k = \frac{0.693}{t_{1/2}} = \frac{0.693}{6 \text{ hrs}} = 0.1155 \text{ h}^{-1}$$

2. Set up the equation for fraction of steady state:

$$\frac{C_p}{C_{p_{ss}}} = 1 - e^{-k \times t}$$

The target is 95%, so $\frac{C_p}{C_{p_{ss}}} = 0.95$

$$0.95 = 1 - e^{-0.1155 \times t}$$

3. Solve for time (t):

$$e^{-0.1155 \times t} = 1 - 0.95$$

$$e^{-0.1155 \times t} = 0.05$$

4. Take the natural logarithm (ln) of both sides:

$$\ln(0.05) = -0.1155 \times t$$

$$-2.9957 = -0.1155 \times t$$

$$t = \frac{-2.9957}{-0.1155} \approx 25.93 \text{ hours}$$

Question 68- A pharmacologist is analyzing the hepatic elimination of Xylocaine® (lidocaine) in a 55-year-old male patient. To accurately assess the drug's metabolism, the extraction ratio must be determined using the patient's specific physiological parameters. The hepatic clearance is 0.9 L/min, hepatic plasma flow is 0.825 L/min and the patient's hematocrit is 45%.

Calculate the hepatic extraction ratio (E) for this patient.

Question 68- Solution: 0.6

1. Convert Hematocrit to a decimal:

$$\text{Hct} = 45\% = 0.45$$

2. Calculate Hepatic Blood Flow (Q): Since clearance is based on whole blood flow in this context, convert plasma flow to blood flow.

$$Q_{\text{blood}} = \frac{Q_{\text{plasma}}}{1 - \text{Hct}}$$

$$Q = \frac{0.825 \text{ L/min}}{1 - 0.45} = \frac{0.825}{0.55} = 1.5 \text{ L/min}$$

3. Calculate Extraction Ratio (E):

$$E = \frac{Cl_H}{Q}$$

$$E = \frac{0.9 \text{ L/min}}{1.5 \text{ L/min}}$$

$$E = 0.6$$

Question 69- A pharmacist receives a prescription to compound an oral suspension for a pediatric patient who cannot swallow pills. The prescription calls for 120 mL of Prevacid® (lansoprazole) 3 mg/mL suspension. The pharmacy currently only stocks Prevacid® (lansoprazole) 30 mg capsules.

Calculate the total number of capsules required to compound the full volume of the suspension.

Question 69- Solution: 12 capsules

1. Calculate the total amount of active ingredient required: Multiply the total volume of the suspension by the prescribed concentration to find the total milligrams of lansoprazole needed.

$$120 \text{ mL} \times 3 \text{ mg/mL} = 360 \text{ mg}$$

2. Calculate the number of capsules needed: Divide the total milligrams required by the strength of the available capsules.

$$\frac{360 \text{ mg}}{30 \text{ mg/capsule}} = 12 \text{ capsules}$$

3. Conclusion: The pharmacist must open 12 capsules of Prevacid® (lansoprazole) and mix the contents with an appropriate vehicle (such as sodium bicarbonate solution or a commercially available suspending agent) to make a final volume of 120 mL.

Question 70- A compounding pharmacist receives a prescription for 12 Tylenol® (acetaminophen) 300 mg suppositories. The base vehicle to be used is cocoa butter. The pharmacist has calibrated the mold and determined the average weight of a blank suppository (base only) is 2.0 g. The density factor of acetaminophen is 1.5.

Calculate the total amount of cocoa butter base required to compound this prescription in grams.

Question 70- Solution: 21.6 g

1. Specific Gravity (Concept):

$$SG = \frac{\text{Weight of substance}}{\text{Weight of equal volume of water}}$$

(Used here as the ratio of drug density to base density)

2. Calculate Total Drug Required:

$$12 \text{ suppositories} \times 300 \text{ mg/supp} = 3600 \text{ mg} = 3.6 \text{ g}$$

3. Calculate Theoretical Base Required (Blank Weight):

$$12 \text{ suppositories} \times 2.0 \text{ g/supp} = 24.0 \text{ g}$$

4. Calculate Base Displaced by Drug:

$$\text{Weight of Base Displaced} = \frac{\text{Total Weight of Drug}}{\text{Density Factor}}$$

Using the density factor (where 1.5 g of drug occupies the same volume as 1.0 g of base):

$$\text{Base Displaced} = \frac{3.6 \text{ g (drug)}}{1.5} = 2.4 \text{ g (base)}$$

5. Calculate Net Base Needed:

Subtract the displaced base from the theoretical total:

$$24.0 \text{ g} - 2.4 \text{ g} = 21.6 \text{ g}$$

Question 71- A pharmacist is compounding fluoxetine 20 mg capsules using lactose as a diluent. Based on the density of the powders, the pharmacist has determined the weight capacity of a Size 1 capsule for each individual ingredient. The weight of fluoxetine powder required to fill a Size 1 capsule is 250 mg and the weight of lactose powder required to fill a Size 1 capsule is 400 mg.

Calculate the amount of lactose, in mg, required to fill one capsule containing the prescribed dose of fluoxetine.

Question 71- Solution: 368 mg

1. Determine the volume fraction occupied by the active drug:

Divide the target dose by the total capacity of the capsule for that specific drug.

$$\text{Fraction}_{\text{drug}} = \frac{\text{Target Dose}}{\text{Capacity}_{\text{drug}}}$$

$$\text{Fraction}_{\text{drug}} = \frac{20 \text{ mg}}{250 \text{ mg}} = 0.08 \text{ (or 8\%)}$$

2. Determine the volume fraction remaining for the diluent:

$$\text{Fraction}_{\text{diluent}} = 1 - \text{Fraction}_{\text{drug}}$$

$$\text{Fraction}_{\text{diluent}} = 1 - 0.08 = 0.92 \text{ (or 92\%)}$$

3. Calculate the weight of lactose needed:

$$\text{Weight}_{\text{lactose}} = \text{Fraction}_{\text{diluent}} \times \text{Capacity}_{\text{lactose}}$$

$$\text{Weight}_{\text{lactose}} = 0.92 \times 400 \text{ mg}$$

$$\text{Weight}_{\text{lactose}} = 368 \text{ mg}$$

Question 72- A hospital pharmacist is preparing 100 mL of a hypertonic infusion for a patient requiring urgent electrolyte replacement. The order requires a specific concentration of potassium chloride dissolved in sterile water for injection. The mass of solute is 2.98 grams. The molecular weight (MW) of potassium chloride is 74.5 g/mol.

Calculate the osmolarity of the prepared solution in mOsm/L.

Question 72- Solution: 800 mOsm/L.

1. Determine the Species Count: Potassium chloride (KCl) is an electrolyte that dissociates into potassium (K^+) and chloride (Cl^-) ions in solution.

$$\text{Species Count} = 2$$

2. Convert Concentration to mg/L: First, determine grams per liter, then convert to milligrams.

$$\text{Concentration} = \frac{2.98 \text{ g}}{100 \text{ mL}} = \frac{29.8 \text{ g}}{1000 \text{ mL}} = 29.8 \text{ g/L}$$

$$29.8 \text{ g/L} \times 1000 \text{ mg/g} = 29,800 \text{ mg/L}$$

3. Calculate Osmolarity: Apply the values to the formula.

$$\text{mOsmol/L} = \frac{\text{Concentration (mg/L)}}{\text{MW}} \times \text{Species Count}$$

$$\text{mOsmol/L} = \frac{29,800 \text{ mg/L}}{74.5 \text{ g/mol}} \times 2$$

$$\text{mOsmol/L} = 400 \times 2 = 800 \text{ mOsmol/L}$$

Question 73- JL is a 62-year-old patient being treated for a severe systemic infection in the ICU. The physician has ordered a continuous IV infusion of vancomycin. The pharmacy team estimates the patient's parameters to determine the appropriate dosing rate to achieve a specific steady-state concentration. The target steady-state plasma concentration is 20 mg/L and the drug clearance is 3.5 L/hr.

Calculate the required rate of infusion in mg/hr.

Question 73- Solution: 70 mg/hr

1. Rearrange Formula:

To solve for the rate of infusion (R_0), rearrange the steady-state equation:

$$C_{p_{ss}} = \frac{F \times R_0}{Cl}$$

$$R_0 = \frac{C_{p_{ss}} \times Cl}{F}$$

2. Calculate:

$$R_0 = \frac{20 \text{ mg/L} \times 3.5 \text{ L/hr}}{1}$$

$$R_0 = 70 \text{ mg/hr}$$

Question 74- A nurse is preparing to administer 100 mL of IV Flagyl® (metronidazole) to a patient for the treatment of an anaerobic bacterial infection. The medication is to be infused over 30 minutes. The pharmacy has sent the medication in a premixed bag, and the nurse must manually regulate the infusion rate using the available tubing. The administration set drop factor (DF) is 10 drops/mL.

Calculate the required flow rate in drops/min.

Question 74- Solution: 33 drops/min

1. Substitute values into the equation:

$$\text{Flow Rate (gtt/min)} = \frac{\text{Volume (mL)} \times \text{Drop Factor (gtt/mL)}}{\text{Time (min)}}$$

$$\text{Flow Rate} = \frac{100 \times 10}{30}$$

$$\text{Flow Rate} = \frac{1000}{30} = 33.33 \text{ drops/min}$$

2. Round: Since partial drops cannot be counted manually, round to the nearest whole number.

$$\text{Flow Rate} \approx 33 \text{ drops/min}$$

Question 75- JS is a 6-year-old female patient (weight: 44 lbs) who has been prescribed amoxicillin oral suspension 20 mg/kg/day divided QID for a dental infection. You only have the 125 mg/5 mL amoxicillin suspension in stock at your pharmacy.

Calculate the volume, in mL, to be administered per individual dose.

Question 75- Solution: 4 mL

1. Convert patient weight to kg:

$$\text{Weight} = \frac{44 \text{ lbs}}{2.2 \text{ lbs/kg}} = 20 \text{ kg}$$

2. Calculate the total daily dose in mg:

$$\text{Daily Dose} = 20 \text{ kg} \times 20 \text{ mg/kg/day} = 400 \text{ mg/day}$$

3. Calculate the individual dose (QID = 4 doses):

$$\text{Dose per administration} = \frac{400 \text{ mg}}{4} = 100 \text{ mg}$$

4. Calculate the volume per dose using proportion:

$$\text{From Concentration Expressions: } \frac{\text{Drug in Dilution}}{\text{Dilution Vol}} = \frac{\text{Drug in Aliquot}}{\text{Aliquot Vol}}$$

$$\frac{125 \text{ mg}}{5 \text{ mL}} = \frac{100 \text{ mg}}{X \text{ mL}}$$

$$X = \frac{100 \text{ mg} \times 5 \text{ mL}}{125 \text{ mg}}$$

$$X = \frac{500}{125} = 4 \text{ mL}$$

Question 76- TR is a 45-year-old male patient (weight: 176 lbs) who presents with a severe acute asthma exacerbation. He is being discharged on a tapering regimen of prednisone oral concentrate 5 mg/mL. The initial prescribed dose is 1mg/kg/day. The tapering instructions state: "maintain initial dose for 2 days, then reduce the daily dose by 20% of the initial dose every 2 days until finished."

Calculate the total cumulative volume (in mL) of the prednisone concentrate the pharmacist must dispense to complete this entire tapering schedule.

Question 76- Solution: 96 mL

1. Convert patient weight to kg:

$$176 \text{ lbs} \div 2.2 \text{ lbs/kg} = 80 \text{ kg}$$

2. Calculate the Initial Daily Dose:

$$80 \text{ kg} \times 1 \text{ mg/kg/day} = 80 \text{ mg/day}$$

3. Calculate the Reduction Step:

The dose reduces by 20% of the initial dose (80 mg).

$$80 \text{ mg} \times 0.20 = 16 \text{ mg}$$

The daily dose decreases by 16 mg every 2 days.

4. Map the tapering intervals (Duration = 2 days per interval):

Interval 1: 80 mg/day

$$\text{Interval 2: } 80 \text{ mg} - 16 \text{ mg} = 64 \text{ mg/day}$$

$$\text{Interval 3: } 64 \text{ mg} - 16 \text{ mg} = 48 \text{ mg/day}$$

$$\text{Interval 4: } 48 \text{ mg} - 16 \text{ mg} = 32 \text{ mg/day}$$

$$\text{Interval 5: } 32 \text{ mg} - 16 \text{ mg} = 16 \text{ mg/day}$$

$$\text{Interval 6: } 16 \text{ mg} - 16 \text{ mg} = 0 \text{ mg (Therapy ends)}$$

5. Calculate Total Cumulative Dose (mg):

Since each interval is 2 days, we sum the daily doses and multiply by 2.

$$\text{Sum of daily doses} = 80 + 64 + 48 + 32 + 16 = 240 \text{ mg}$$

$$\text{Total Cumulative Dose} = 240 \text{ mg} \times 2 \text{ days} = 480 \text{ mg}$$

6. Convert Total Mass to Volume:

Using the concentration of 5 mg/mL:

$$\text{Volume} = \frac{480 \text{ mg}}{5 \text{ mg/mL}} = 96 \text{ mL}$$

Question 77- JR is a 52-year-old male (height: 6 feet, weight: 216 lbs) who presents to the pharmacy clinic to discuss weight loss options. The physician is considering prescribing Xenical® (orlistat) and requests a baseline assessment of the patient's body composition.

Calculate the patient's Body Mass Index (BMI) in kg/m^2 .

Question 77- Solution: 29.4 kg/m^2

1. Convert Weight to kg:

$$\text{Weight} = \frac{216 \text{ lbs}}{2.2 \text{ lbs/kg}} \approx 98.18 \text{ kg}$$

2. Convert Height to meters:

$$6 \text{ ft} = 72 \text{ inches}$$

$$72 \text{ in} \times 2.54 \text{ cm/in} = 182.88 \text{ cm}$$

$$182.88 \text{ cm} \div 100 = 1.8288 \text{ m}$$

3. Calculate BMI:

$$\text{BMI} = \frac{\text{Weight (kg)}}{(\text{Height (m)})^2}$$

$$\text{BMI} = \frac{98.18 \text{ kg}}{(1.8288 \text{ m})^2}$$

$$\text{BMI} = \frac{98.18}{3.3445} = 29.35$$

$$\text{BMI} \approx 29.4 \text{ kg/m}^2$$

Note: Based on the BMI table, this patient falls into the 'Overweight' category (25 – 29.9).

Question 78- A pharmacist receives a prescription to compound 50 mL of a 1% ophthalmic solution using Isopto® Carpine (pilocarpine hydrochloride). The pharmacist must calculate the amount of sodium chloride required to render the final solution isotonic with lacrimal fluid. The molecular weight of pilocarpine hydrochloride is 245 g/mol, the dissociation factor (i) of pilocarpine hydrochloride is 1.8, the molecular weight of sodium chloride (NaCl) is 58.5 g/mol, the dissociation factor (i) of sodium chloride is 1.8 and the isotonic concentration of sodium chloride is 0.9%.

Calculate the amount of sodium chloride (in grams) to be added to the solution to achieve isotonicity.

Question 78- Solution: 0.33 g

1. Sodium Chloride Equivalent (E):

$$E = \frac{58.5}{i_{\text{NaCl}}} \times \frac{i_{\text{drug}}}{\text{MW}_{\text{drug}}}$$

2. Amount of Drug:

$$\text{Mass} = \text{Volume} \times \text{Concentration \%}$$

3. Total NaCl Required:

$$\text{Total NaCl} = \text{Volume} \times 0.9\%$$

4. NaCl to Add:

$$\text{NaCl}_{\text{add}} = \text{Total NaCl} - (\text{Drug Mass} \times E)$$

5. Calculate the Sodium Chloride Equivalent (E-value) for pilocarpine hydrochloride:

$$E = \frac{58.5}{1.8} \times \frac{1.8}{245} = 32.5 \times 0.00735 \approx 0.239$$

6. Calculate the amount of pilocarpine hydrochloride in the prescription:

$$50 \text{ mL} \times 1\% = 50 \times 0.01 = 0.5 \text{ g}$$

7. Calculate the sodium chloride equivalent of the drug present:

$$0.5 \text{ g (drug)} \times 0.239 \text{ (E-value)} = 0.1195 \text{ g (NaCl equivalents)}$$

8. Calculate the total amount of sodium chloride needed for 50 mL of isotonic solution:

$$50 \text{ mL} \times 0.9\% = 50 \times 0.009 = 0.45 \text{ g}$$

9. Calculate the amount of sodium chloride to add:

$$0.45 \text{ g} - 0.1195 \text{ g} = 0.3305 \text{ g}$$

10. Final Answer:

Rounding to two decimal places, add 0.33 g of sodium chloride.

Question 79- A compounding pharmacist is preparing a buffer solution to stabilize the formulation of Motrin® (ibuprofen) suspension. To ensure the drug remains in solution, the pharmacist creates a mixture of a weak acid and its conjugate base. The concentration of the weak acid is 0.15 M, the concentration of the conjugate base is 0.25 M and the acid dissociation constant (pKa) is 4.85.

Calculate the pH of the buffer solution.

Question 79- Solution: 5.07

1. Substitute values into the equation:

$$\text{pH} = \text{pKa} + \log\left(\frac{[\text{A}^-]}{[\text{HA}]}\right)$$

$$\text{pH} = 4.85 + \log\left(\frac{0.25}{0.15}\right)$$

2. Solve the logarithmic term:

$$\frac{0.25}{0.15} \approx 1.667$$

$$\log(1.667) \approx 0.2218$$

3. Calculate Final pH:

$$\text{pH} = 4.85 + 0.2218 = 5.0718 \approx 5.07$$

Question 80- JL is a 65-year-old male admitted for the treatment of severe hypokalemia. The physician orders an intravenous infusion of potassium chloride. The order specifies a dose of 1490 mg. The molecular weight of potassium chloride (KCl) is 74.5 g/mol.

Calculate the dose of potassium chloride in milliequivalents (mEq).

Question 80- Solution: 20 mEq

1. Identify Variables:

$$\text{Mass} = 1490 \text{ mg}$$

$$\text{Molecular Weight (MW)} = 74.5 \text{ g/mol}$$

$$\text{Valence} = 1 \text{ (since } K^+ \text{ has a charge of } +1\text{)}$$

2. Calculate:

$$\text{mEq} = \frac{\text{Mass (mg)} \times \text{Valence}}{\text{MW}}$$

$$\text{mEq} = \frac{1490 \times 1}{74.5}$$

$$\text{mEq} = 20 \text{ mEq}$$

Question 81- A nuclear pharmacist prepares a unit dose of Cardiolite® (technetium 99m sestamibi), which has a half-life of 6 hours, for a myocardial perfusion imaging study. The dose was assayed earlier in the morning, and the pharmacist must calculate the remaining radioactivity at the scheduled administration time to ensure the patient receives the correct amount of radiation. The initial activity (A_0) is 900 MBq. The calibration time (t_0) is 07:00. The administration time (t) is 10:00.

Calculate the remaining activity (A_t) at the time of administration in MBq.

Question 81- Solution: 636.3 MBq

1. Determine the time interval (t):

$$t = 10:00 - 07:00 = 3 \text{ hours}$$

2. Calculate the decay constant (k):

Using the formula from the Elimination Kinetics section:

$$k = \frac{0.693}{t_{1/2}} \text{ (Decay constant calculation)}$$

$$k = \frac{0.693}{6 \text{ h}} = 0.1155 \text{ h}^{-1}$$

3. Calculate the remaining activity (A_t):

Using the One Compartment Model equation adapted for radioactive decay:

$$C = C_0 \times e^{-k \times t} \text{ (First-order decay equation, where C represents Activity A)}$$

$$A_t = 900 \times e^{-(0.1155 \times 3)}$$

$$A_t = 900 \times e^{-0.3465}$$

$$A_t = 900 \times 0.7071$$

$$A_t \approx 636.3 \text{ MBq}$$

Question 82- A clinical pharmacokinetic study is conducted to determine the absolute bioavailability of Cipro® (ciprofloxacin). A healthy volunteer receives an intravenous bolus dose of 400 mg. Then, after an appropriate washout period, they receive an oral tablet dose of 500 mg. The area under the curve (AUC) after both doses is 20 mg*hr/L.

Calculate the absolute bioavailability (F) of the oral ciprofloxacin formulation expressed as a percentage.

Question 82- Solution: 80%

1. Substitute the given values:

$$F = \frac{AUC_{oral} \times D_{IV}}{AUC_{IV} \times D_{oral}}$$

$$F = \frac{20 \text{ mg} \cdot \text{h/L} \times 400 \text{ mg}}{20 \text{ mg} \cdot \text{h/L} \times 500 \text{ mg}} = 0.8$$

2. Convert to percentage:

$$0.8 \times 100 = 80\%$$

Question 83- BT is a 55-year-old female patient (height: 65 inches, weight: 90 kg) who has colorectal cancer. She is scheduled to receive a bolus dose of fluorouracil 6 mg/kg as part of her chemotherapy regimen. Due to her weight, the oncologist specifies that the dose must be calculated using her Ideal Body Weight (IBW).

Calculate the required dose of fluorouracil, in mg.

Question 83- Solution: 342 mg

1. Calculate Ideal Body Weight (IBW):

$$\text{IBW} = 45.5\text{ kg} + 2.3 \text{ kg} \times (\text{Height (inches)} - 60)$$

$$\text{IBW} = 45.5 \text{ kg} + 2.3 \text{ kg} \times (65 - 60 \text{ inches})$$

$$\text{IBW} = 45.5 \text{ kg} + 2.3 \text{ kg} \times (5 \text{ inches})$$

$$\text{IBW} = 45.5\text{ kg} + 11.5 \text{ kg} = 57 \text{ kg}$$

2. Calculate Dose:

$$\text{Dose} = 57 \text{ kg} \times 6 \text{ mg/kg}$$

$$\text{Dose} = 342 \text{ mg}$$

Question 84- A pharmacy technician is verifying the weight of raw ingredients for a compound suspension containing Tylenol® (acetaminophen). The master formulation record calls for a precise amount of the active pharmaceutical ingredient, but the actual amount weighed on the balance deviates slightly from the target. The desired quantity was 1250 mg, whereas the actual quantity weighed was 1210 mg.

Calculate the percentage of error of the weighed ingredient.

Question 84- Solution: 3.2%

1. Formulas:

$$\% \text{ Error} = \frac{|\text{Actual Value} - \text{Theoretical Value}|}{\text{Theoretical Value}} \times 100\%$$

2. Calculate the absolute difference:

$$|1210 - 1250| = |-40| = 40 \text{ mg}$$

3. Divide by the theoretical value:

$$\frac{40 \text{ mg}}{1250 \text{ mg}} = 0.032$$

4. Convert to percentage:

$$0.032 \times 100 = 3.2\%$$

Question 85- A 72-year-old male patient (weight: 80 kg, height: 70 inches, SCr: 1.2 mg/dL) is admitted to the ICU with a severe infection. The medical team decides to administer vancomycin via continuous IV infusion to maintain a target steady-state concentration of 20 mg/L. The clinical pharmacist must determine the appropriate infusion rate based on the patient's renal function.

Calculate the required IV infusion rate (R_0) in mg/hr, assuming the patient's total drug clearance (Cl) is equal to 80% of their calculated Creatinine Clearance (CrCl) using Ideal Body Weight (IBW).

Question 85- Solution: 55.2 mg/hr

1. Calculate Ideal Body Weight (IBW):

Since the patient is male and 70 inches tall:

$$\text{IBW} = 50\text{kg} + 2.3\text{kg} \times (\text{Height (inches)} - 60)$$

$$\text{IBW} = 50 + 2.3(70 - 60) = 50 + 2.3(10) = 73 \text{ kg}$$

2. Calculate Creatinine Clearance (CrCl):

Using IBW (73 kg), Age (72), and SCr (1.2 mg/dL):

$$\text{CrCl (mL/min)} = \frac{(140 - \text{Age}) \times \text{Weight (kg)}}{72 \times \text{SCr (mg/dL)}}$$

$$\text{CrCl} = \frac{(140 - 72) \times 73}{72 \times 1.2}$$

$$\text{CrCl} = \frac{68 \times 73}{86.4} = \frac{4964}{86.4} \approx 57.45 \text{ mL/min}$$

3. Determine Drug Clearance (Cl):

The problem states Cl is 80% of CrCl.

$$\text{Cl} = 0.80 \times 57.45 \text{ mL/min} = 45.96 \text{ mL/min}$$

Convert Cl to L/hr to match the target concentration units (mg/L) and requested rate units (mg/hr):

$$Cl = 45.96 \frac{\text{mL}}{\text{min}} \times \frac{60 \text{ min}}{1 \text{ hr}} \times \frac{1 \text{ L}}{1000 \text{ mL}} = 2.758 \text{ L/hr}$$

4. Calculate Infusion Rate (R_0):

Rearranging the steady-state equation for R_0 (assuming IV administration, Bioavailability $F = 1$):

$$Cp_{ss} = \frac{F \times R_0}{Cl}$$

$$R_0 = Cp_{ss} \times Cl$$

$$R_0 = 20 \text{ mg/L} \times 2.758 \text{ L/hr}$$

$$R_0 = 55.16 \text{ mg/hr}$$

Rounding to one decimal place:

$$R_0 \approx 55.2 \text{ mg/hr}$$

Question 86- JL is a 55-year-old male patient (weight: 80 kg) receiving drug therapy for epilepsy. He is currently taking Dilantin® (phenytoin sodium) 300 mg PO daily. His seizure control is inadequate. A steady-state trough level was drawn and was found to be 7 mg/L. The pharmacy team intends to increase his dose to achieve a therapeutic target concentration of 15 mg/L. The population Michaelis-Menten constant (K_m) is 4 mg/L. The bioavailability of phenytoin sodium is 100% and the salt factor is 0.92.

Calculate the new daily dose of Dilantin® required to achieve the target concentration. Report your answer rounded to the nearest whole milligram.

Question 86- Solution: 372 mg

1. Calculate the current effective daily rate (R_a) of active drug:

$$\frac{S \times F \times D}{\tau} = \frac{V_{max} \times Cp_{ss}}{K_m + Cp_{ss}}$$

$$R_a = \frac{S \times F \times D}{\tau} = \frac{0.92 \times 1 \times 300 \text{ mg}}{1 \text{ day}} = 276 \text{ mg/day}$$

2. Calculate the patient's maximum metabolic rate (V_{max}) using current data:

$$V_{\max} = \frac{\text{Daily Rate} \times (K_m + C_{p_{ss}})}{C_{p_{ss}}}$$

$$V_{\max} = \frac{276 \times (4 + 7)}{7}$$

$$V_{\max} = \frac{276 \times 11}{7} = \frac{3036}{7} \approx 433.71 \text{ mg/day}$$

3. Calculate the required rate of active drug to achieve Target $C_{p_{ss}}$ of 15 mg/L:

$$\text{Rate}_{\text{target}} = \frac{V_{\max} \times C_{p_{ss, \text{target}}}}{K_m + C_{p_{ss, \text{target}}}}$$

$$\text{Rate}_{\text{target}} = \frac{433.71 \times 15}{4 + 15}$$

$$\text{Rate}_{\text{target}} = \frac{6505.65}{19} \approx 342.40 \text{ mg/day}$$

4. Convert the active rate back to the capsule dose (accounting for S):

$$D_{\text{new}} = \frac{\text{Rate}_{\text{target}}}{S \times F}$$

$$D_{\text{new}} = \frac{342.40}{0.92 \times 1} \approx 372 \text{ mg}$$

Question 87- RJ is a 54-year-old male being treated for complex partial seizures with Dilantin® (phenytoin). Due to the non-linear pharmacokinetics of the drug, dose adjustments have been challenging. The patient has provided steady-state plasma concentration data from two different dosage regimens. The clinician aims to target a specific steady-state concentration of 15 mg/L. The first regimen was 300 mg/day and achieved a steady-state concentration of 7.5 mg/L. The second regimen was 400 mg/day and achieved a steady-state concentration of 20 mg/L. The bioavailability and salt factor are both 1.

Calculate the required daily dose to achieve the target concentration of 15 mg/L after first determining the patient-specific Michaelis-Menten parameters ($V_{\max}$ and K_m).

Question 87- Solution: 375 mg/day

1. Daily Rate (Michaelis-Menten):

$$R_a = \frac{V_{\max} \times C_{p_{ss}}}{K_m + C_{p_{ss}}}$$

(assuming $F = 1, S = 1$)

2. Linearized form for parameter estimation:

$$R_a = V_{\max} - K_m \times \left(\frac{R_a}{C_{p_{ss}}} \right)$$

3. Direct calculation for K_m (Two-point method):

$$K_m = \frac{R_{a2} - R_{a1}}{\left(\frac{R_{a1}}{C_{p_{ss1}}} \right) - \left(\frac{R_{a2}}{C_{p_{ss2}}} \right)}$$

4. Calculate Clearance rates for each point (R/C_p):

$$\text{Point 1: } \frac{R_{a1}}{C_{p_{ss1}}} = \frac{300}{7.5} = 40 \text{ L/day}$$

$$\text{Point 2: } \frac{R_{a2}}{C_{p_{ss2}}} = \frac{400}{20} = 20 \text{ L/day}$$

5. Calculate K_m (Michaelis-Menten constant):

Using the slope formula derived from the linearized equation:

$$K_m = \frac{400 - 300}{40 - 20} = \frac{100}{20} = 5 \text{ mg/L}$$

6. Calculate $V_{\max}$ (Maximum metabolic rate):

Using Regimen B data and the calculated K_m :

$$R_{a2} = \frac{V_{\max} \times C_{p_{ss2}}}{K_m + C_{p_{ss2}}}$$

$$400 = \frac{V_{\max} \times 20}{5 + 20}$$

$$400 = \frac{20V_{\max}}{25}$$

$$10000 = 20V_{\max}$$

$$V_{\max} = 500 \text{ mg/day}$$

7. Calculate Target Dose (R_{target}):

Now plug $V_{\max}$, K_m , and the target $C_{p_{ss}}$ (15 mg/L) into the main equation:

$$R_{\text{target}} = \frac{V_{\max} \times C_{p_{\text{target}}}}{K_m + C_{p_{\text{target}}}}$$

$$R_{\text{target}} = \frac{500 \times 15}{5 + 15}$$

$$R_{\text{target}} = \frac{7500}{20}$$

$$R_{\text{target}} = 375 \text{ mg/day}$$

Question 88- LK is a 58-year-old male patient (weight: 75 kg) being treated for severe gram-negative sepsis. He receives a 500 mg dose of amikacin administered as a 1-hour intravenous infusion. To prevent toxicity and ensure efficacy, the clinical pharmacist requests serum levels to calculate the patient's specific pharmacokinetic parameters. One hour after the infusion ended, the plasma concentration was 28 mg/L. Five hours after the infusion ended, the plasma concentration was 10 mg/L.

Calculate the patient's total clearance (Cl) in L/h.

Question 88- Solution: 3.13 L/h

1. Elimination Rate Constant:

$$k = \frac{\ln(C_{p1}) - \ln(C_{p2})}{t_2 - t_1}$$

Determine the time difference between the two measurements.

$$t_{\Delta} = 5 \text{ hours} - 1 \text{ hour} = 4 \text{ hours}$$

$$k = \frac{\ln(28) - \ln(10)}{4}$$

$$k = \frac{3.332 - 2.302}{4} = \frac{1.03}{4} = 0.2575 \text{ h}^{-1}$$

2. Concentration Decay (to find peak at end of infusion):

$$C_{p_t} = C_{p_0} \times e^{-k \times t} \text{ (Rearranged to solve for } C_{p_0})$$

The first level (28 mg/L) was drawn 1 hour after the infusion stopped. We must back-extrapolate to time 0 (the moment the infusion ended) to find the true peak required for the infusion formula.

$$28 = C_{\max} \times e^{-0.2575 \times 1}$$

$$C_{\max} = \frac{28}{e^{-0.2575}} = \frac{28}{0.773} = 36.22 \text{ mg/L}$$

3. IV Infusion (during infusion):

$$C_p = \frac{F \times R_0}{Cl} \times (1 - e^{-k \times t})$$

(Rearranged to solve for Cl)

Use the IV infusion equation where time t is the duration of the infusion (1 hour), F = 1 (IV), and C_p is the C_{max} calculated in step 2.

$$R_0 = \frac{500 \text{ mg}}{1 \text{ h}} = 500 \text{ mg/h}$$

$$36.22 = \frac{1 \times 500}{Cl} \times (1 - e^{-0.2575 \times 1})$$

$$36.22 = \frac{500}{Cl} \times (1 - 0.773)$$

$$36.22 = \frac{500}{Cl} \times (0.227)$$

$$Cl = \frac{500 \times 0.227}{36.22} = \frac{113.5}{36.22}$$

$$Cl \approx 3.13 \text{ L/h}$$

Question 89- TR is a 55-year-old male (weight: 78 kg) undergoing treatment for complex partial seizures. He has been receiving maintenance therapy with Dilantin® (phenytoin). Due to non-linear pharmacokinetics, the pharmacist must determine the patient's specific metabolic capacity before increasing the dose further. To determine this, the pharmacist asks the patient to adhere to two separate dosing regimens for one week each, and steady-state serum concentrations were obtained for both.

Labs:

Regimen A: 300 mg/day resulted in $C_{p_{ss}} = 8$ mg/L

Regimen B: 450 mg/day resulted in $C_{p_{ss}} = 24$ mg/L

Assume bioavailability (F) and salt factor (S) are equivalent to 1 for the purpose of this calculation.

Calculate the patient's maximum metabolic rate (V_{max}) in mg/day.

Question 89- Solution: 600 mg/day

1. Set up the simultaneous equations:

$$\frac{F \times D}{\tau} = \frac{V_{max} \times C_{p_{ss}}}{K_m + C_{p_{ss}}}$$

$$\text{Equation 1: } 300 = \frac{V_{max} \times 8}{K_m + 8}$$

$$\text{Equation 2: } 450 = \frac{V_{max} \times 24}{K_m + 24}$$

2. Rearrange to isolate V_{max} terms:

$$\text{Eq 1: } 300(K_m + 8) = 8V_{max} \Rightarrow 300K_m + 2400 = 8V_{max}$$

$$\text{Eq 2: } 450(K_m + 24) = 24V_{max} \Rightarrow 450K_m + 10800 = 24V_{max}$$

3. Solve for K_m :

Multiply Eq 1 by 3 to align the V_{max} term with Eq 2:

$$3 \times (300K_m + 2400) = 3 \times (8V_{max})$$

$$900K_m + 7200 = 24V_{max}$$

Now set the modified Eq 1 equal to Eq 2:

$$900K_m + 7200 = 450K_m + 10800$$

$$450K_m = 3600$$

$$K_m = \frac{3600}{450} = 8 \text{ mg/L}$$

4. Solve for $V_{\max}$:

Substitute $K_m = 8$ back into the original Equation 1:

$$300 = \frac{V_{\max} \times 8}{8 + 8}$$

$$300 = \frac{8V_{\max}}{16}$$

$$300 \times 16 = 8V_{\max}$$

$$4800 = 8V_{\max}$$

$$V_{\max} = \frac{4800}{8} = 600 \text{ mg/day}$$

Question 90- A 65-year-old male patient is receiving a 400 mg IV bolus dose of gentamicin for the treatment of endocarditis. To establish the patient's specific pharmacokinetic parameters for future dosing, plasma concentrations were measured at two specific time points. The plasma concentration at 3 hours (C_{p1}) was 10 mg/L, whereas the plasma concentration at 9 hours (C_{p2}) was 2.5 mg/L.

Calculate the patient's total clearance (Cl) in L/h.

Question 90- Solution: 4.62 L/h

1. Elimination Rate Constant:

$$k = \frac{\ln(C_{p1}) - \ln(C_{p2})}{t_2 - t_1}$$

$$k = \frac{\ln(10) - \ln(2.5)}{9 - 3}$$

$$k = \frac{2.302 - 0.916}{6} = \frac{1.386}{6} = 0.231 \text{ h}^{-1}$$

2. Calculate the initial concentration (C_0):

$$C_t = C_{p0} \times e^{-k \times t}$$

$$C_0 = \frac{C_t}{e^{-k \times t}} = \frac{10 \frac{\text{mg}}{\text{L}}}{e^{-0.231 \text{ h}^{-1} \times 3 \text{ hrs}}} = 20 \frac{\text{g}}{\text{L}}$$

3. Calculate the Volume of Distribution (Vd):

$$V_d = \frac{\text{Dose}}{C_0} = \frac{400 \text{ mg}}{20 \text{ mg/L}} = 20 \text{ L}$$

4. Calculate Total Clearance (Cl):

$$Cl = k \times V_d = 0.231 \text{ h}^{-1} \times 20 \text{ L}$$

$$Cl = 4.62 \text{ L/h}$$

Question 91- JL is a 68-year-old male receiving a 160 mg IV bolus dose of gentamicin for suspected sepsis. To tailor his ongoing therapy, two plasma concentration samples were drawn to assess his specific pharmacokinetic parameters, assuming first-order elimination. The plasma concentration at 2 hours post-dose was 6 mg/L, whereas the plasma concentration at 8 hours post-dose was 2 mg/L.

Calculate the patient's Volume of Distribution (V_d) in liters.

Question 91- Solution: 18.5 L

1. Elimination Rate Constant:

$$k = \frac{\ln(C_{p1}) - \ln(C_{p2})}{t_2 - t_1}$$

$$k = \frac{\ln(6.0) - \ln(2.0)}{8 - 2}$$

$$k = \frac{1.7918 - 0.6931}{6}$$

$$k = \frac{1.0987}{6} = 0.183 \text{ h}^{-1}$$

2. Calculate the initial concentration (C_{p0}) using t₁:

$$C_p = C_{p0} \times e^{-k \times t} \text{ (Rearranged to solve for } C_{p0})$$

$$6.0 \text{ mg/L} = C_{p0} \times e^{-(0.183 \text{ h}^{-1} \times 2 \text{ hrs})}$$

$$C_{p0} = \frac{6.0}{0.6934} = 8.65 \text{ mg/L}$$

3. Calculate Volume of Distribution (V_d):

$$V_d = \frac{Ab}{C_p} \rightarrow V_d = \frac{\text{Dose}}{C_{p0}} \text{ (for IV Bolus)}$$

$$V_d = \frac{\text{Dose}}{C_{p0}}$$

$$V_d = \frac{160 \text{ mg}}{8.65 \text{ mg/L}} = 18.5 \text{ L}$$

Question 92- A 72-year-old female patient (height: 65 inches, weight: 82 kg, SCr: 1.4 mg/dL) is admitted for a severe gram-negative infection. She requires treatment with gentamicin. The clinical pharmacist needs to calculate a maintenance dose to be administered intravenously every 12 hours to achieve a target average steady-state plasma concentration of 5 mg/L. Assume the drug's clearance is equal to the patient's creatinine clearance and bioavailability is 100%.

Calculate the required maintenance dose in milligrams (mg), rounded to the nearest whole number.

Question 92- Solution: 138 mg

1. Calculate Ideal Body Weight (IBW):

$$IBW_{\text{female}} = 45.5 \text{ kg} + 2.3 \text{ kg} \times (\text{Height (inches)} - 60)$$

$$IBW = 45.5 + 2.3 \times (65 - 60)$$

$$IBW = 45.5 + 2.3 \times 5 = 45.5 + 11.5 = 57 \text{ kg}$$

2. Decide which weight to use:

$$IBW < ABW, \text{ so we will not use ABW}$$

$$130\% \text{ of IBW} = 1.3 \times 57 \text{ kg} = 74.1 \text{ kg}$$

$$130\% \text{ of IBW} < ABW, \text{ we will use AdjBW}$$

3. Calculate adjusted body weight:

$$\text{AdjBW} = IBW + 0.4 \times (ABW - IBW)$$

$$\text{AdjBW} = 57 \text{ kg} + 0.4 \times (82 \text{ kg} - 57 \text{ kg})$$

$$\text{AdjBW} = 67 \text{ kg}$$

4. Calculate Clearance (Cl):

$$\text{CrCl}_{\text{female}} = \frac{(140 - \text{Age}) \times \text{Weight (kg)}}{72 \times \text{SCr (mg/dL)}} \times 0.85$$

$$\text{Cl} = \frac{(140 - 72) \times 67 \text{ kg}}{72 \times 1.4} \times 0.85 = 38.4 \text{ mL/min}$$

5. Convert Clearance to L/h:

$$38.4 \text{ mL/min} \times \frac{60 \text{ min}}{1 \text{ hr}} \times \frac{1 \text{ L}}{1000 \text{ mL}} = 2.3 \text{ L/h}$$

6. Calculate Maintenance Dose (D):

Rearrange the average concentration formula to solve for D:

$$\text{Cp}_{\text{avg,ss}} = \frac{F \times D}{\text{Cl} \times \tau}$$

$$D = \frac{\text{Cp}_{\text{avg,ss}} \times \text{Cl} \times \tau}{F}$$

$$D = \frac{5 \text{ mg/L} \times 2.3 \text{ L/h} \times 12 \text{ h}}{1}$$

$$D = 138 \text{ mg}$$

Question 93- JL is a 55-year-old male currently receiving 300 mg/day of Dilantin® (phenytoin) for seizure control. His steady-state plasma concentration (Cp_{ss}) was measured at 7 mg/L. The clinician wishes to increase the dose to achieve a target steady-state concentration of 15 mg/L. Assume the patient's absorption is complete ($F = 1$) and follows Michaelis-Menten kinetics. The Michaelis-Menten constant (K_m) is 5 mg/L.

Calculate the new daily maintenance dose required to achieve the target concentration, rounded to the nearest whole mg.

Question 93- Solution: 386 mg/day

1. Daily Rate (Michaelis-Menten):

$$\frac{F \times D}{\tau} = \frac{V_{\max} \times C_{p_{ss}}}{K_m + C_{p_{ss}}}$$

2. Calculate $V_{\max}$:

$$V_{\max} = \frac{R_a \times F \times (K_m + C_{p_{ss}})}{C_{p_{ss}}} = \frac{300 \frac{\text{mg}}{\text{day}} \times 1 \times (5 \frac{\text{mg}}{\text{L}} + 7 \frac{\text{mg}}{\text{L}})}{7 \text{ mg/L}} = 514.3 \text{ mg/day}$$

3. Calculate the new dose:

Now, substitute the calculated $V_{\max}$ and the target concentration (15 mg/L) back into the daily rate equation.

$$\frac{F \times D}{\tau} = \frac{V_{\max} \times C_{p_{ss}}}{K_m + C_{p_{ss}}}$$

$$\frac{1 \times D}{1} = \frac{514.3 \text{ mg/day} \times 15 \text{ mg/L}}{5 \text{ mg/L} + 15 \text{ mg/L}}$$

$$D = 386 \text{ mg/day}$$

Question 94- TR is a 68-year-old male admitted to the cardiac care unit for severe atrial fibrillation. He is administered a rapid IV bolus of digoxin. The pharmacokinetics of the drug in this patient are best described by a two-compartment model. The distribution coefficient (C_1) is 12.5 ng/mL. The distribution rate constant (α) is 4.5 h⁻¹. The elimination coefficient (C_2) is 1.8 ng/mL. The elimination rate constant (β) is 0.15 h⁻¹.

Calculate the plasma concentration of the drug 30 minutes after the bolus dose is administered.

Question 94- Solution: 3.0 ng/mL

1. Convert Time Units:

The rate constants are in h⁻¹, so time must be in hours.

$$t = 30 \text{ minutes} = 0.5 \text{ hours.}$$

2. Calculate Distribution Phase ($C_1 \times e^{-\alpha \times t}$):

$$C_1 \times e^{-\alpha \times t} = 12.5 \times e^{-(4.5 \times 0.5)} \approx 1.3175 \text{ ng/mL}$$

3. Calculate Elimination Phase ($C_2 \times e^{-\beta \times t}$):

$$C_2 \times e^{-\beta \times t} = 1.8 \times e^{-(0.15 \times 0.5)} \approx 1.6699 \text{ ng/mL}$$

4. Sum the Terms:

$$C = C_1 \times e^{-\alpha \times t} + C_2 \times e^{-\beta \times t}$$

$$C_t = 1.3175 + 1.6699 = 2.9874 \text{ ng/mL}$$

5. Note on Rounding:

Depending on calculator precision for e, values may range slightly.

$$C_p \approx 3.0 \text{ ng/mL}$$

Question 95- A 68-year-old male patient with congestive heart failure is being stabilized on digoxin. The clinical team wishes to establish a dosing regimen that maintains the plasma concentration within a strict therapeutic window. Digoxin has a half-life of 36 hours. The target steady-state peak is 2.0 ng/mL and the target steady-state trough is 0.8 ng/mL.

Calculate the maximum dosing interval in hours that maintains the serum concentration within these specific peak and trough limits (round to the nearest whole number).

Question 95- Solution: 48 hours

1. Calculate the Elimination Rate Constant (k):

Rearrange the half-life equation to solve for k.

$$t_{1/2} = \frac{0.693}{k}$$

$$k = \frac{0.693}{t_{1/2}}$$

$$k = \frac{0.693}{36 \text{ h}} \approx 0.01925 \text{ h}^{-1}$$

2. Calculate τ :

$$\frac{C_{p_{\max,ss}}}{C_{p_{\min,ss}}} = \frac{1}{e^{-k \times \tau}}$$

$$\frac{2.0 \text{ ng/mL}}{0.8 \text{ ng/mL}} = \frac{1}{e^{-0.01925 \text{ hr}^{-1} \times \tau}}$$

$$2.5 = \frac{1}{e^{-0.01925 \text{ hr}^{-1} \times \tau}}$$

$$\frac{1}{2.5} = e^{-0.01925 \text{ hr}^{-1} \times \tau}$$

$$0.4 = e^{-0.01925 \text{ hr}^{-1} \times \tau}$$

Use the natural logarithm rule $\rightarrow \ln(e^x) = x$

$$\ln(0.4) = -0.01925 \text{ hr}^{-1} \times \tau$$

$$\tau = \frac{\ln(0.4)}{-0.01925} = 47.6 \text{ hrs}$$

3. Final Rounding:

The calculated interval is approximately 47.6 hours. To ensure the fluctuation does not exceed the limit (keeping the trough from dropping too low or peak going too high given a fixed dose), standard dosing would look for the nearest practical interval.

$$\tau \approx 48 \text{ hours}$$

Question 96- A 45-year-old male is prescribed a tapering regimen of prednisone. The pharmacy receives the following written prescription: "Prednisone 10 mg scored tablets. Take 60 mg PO daily for 3 days; then reduce dose by 10 mg every 2 days until the daily dose is 20 mg. Then take 20 mg daily for 4 days, then 10 mg daily for 4 days, then 5 mg daily for 4 days, then stop."

Calculate the total cumulative dose (in mg) the patient will consume over the entire course of therapy.

Question 96- Solution: 560 mg

1. Phase 1 (Initial High Dose):

$$60 \text{ mg/day} \times 3 \text{ days} = 180 \text{ mg}$$

2. Phase 2 (The Taper):

Instruction: Reduce by 10 mg every 2 days until reaching 20 mg.

$$\text{Step A: } 60 \text{ mg} - 10 \text{ mg} = 50 \text{ mg.}$$

$$50 \text{ mg/day} \times 2 \text{ days} = 100 \text{ mg}$$

$$\text{Step B: } 50 \text{ mg} - 10 \text{ mg} = 40 \text{ mg.}$$

$$40 \text{ mg/day} \times 2 \text{ days} = 80 \text{ mg}$$

$$\text{Step C: } 40 \text{ mg} - 10 \text{ mg} = 30 \text{ mg.}$$

$$30 \text{ mg/day} \times 2 \text{ days} = 60 \text{ mg}$$

Step D: The next reduction would be 20 mg. The specific instructions for the 20 mg tier now take over.

3. Phase 3 (Specific Instructions): Instruction: 20 mg daily for 4 days.

$$20 \text{ mg/day} \times 4 \text{ days} = 80 \text{ mg}$$

4. Phase 4 (Specific Instructions): Instruction: 10 mg daily for 4 days.

$$10 \text{ mg/day} \times 4 \text{ days} = 40 \text{ mg}$$

5. Phase 5 (Final Taper):

Instruction: 5 mg daily for 4 days.

$$5 \text{ mg/day} \times 4 \text{ days} = 20 \text{ mg}$$

6. Calculate Sum:

$$180 \text{ mg} + 100 \text{ mg} + 80 \text{ mg} + 60 \text{ mg} + 80 \text{ mg} + 40 \text{ mg} + 20 \text{ mg} = 560 \text{ mg}$$

Question 97- JR is a 72-year-old male patient (weight: 70 kg, height: 178 cm, SCr: 1.4 mg/dL) admitted for the management of atrial fibrillation. He is being started on digoxin. The clinical pharmacist needs to estimate the drug's half-life to determine the time to reach steady-state. The volume of distribution is 7 L/kg. It is assumed that digoxin clearance ~ creatinine clearance.

Calculate the estimated elimination half-life ($t_{1/2}$) of the drug, in hours.

Question 97- Solution: 120 hours

1. Calculate IBW:

$$\text{IBW} = 50 \text{ kg} + 2.3 (\# \text{ of inches over } 5 \text{ feet} (= 60 \text{ inches}))$$

$$\frac{178 \text{ cm}}{2.54 \frac{\text{cm}}{\text{inch}}} = 70 \text{ inches}$$

$$\text{IBW} = 50 \text{ kg} + (2.3 \text{ kg} \times 10 \text{ inches}) = 73 \text{ kg}$$

since ABW (70 kg) < IBW (73 kg), we will use ABW

2. Creatinine Clearance:

$$\text{CrCl (mL/min)} = \frac{(140 - \text{Age}) \times \text{Weight (kg)}}{72 \times \text{SCr (mg/dL)}}$$

$$\text{CrCl} = \frac{(140 - 72) \times 70 \text{ kg}}{72 \times 1.4 \text{ mg/dL}} = 47.2 \text{ mL/min}$$

3. Convert Clearance to L/h:

Since Vd is in Liters and we need $t_{1/2}$ in hours, convert Cl to consistent units.

$$\text{Cl} = 47.22 \frac{\text{mL}}{\text{min}} \times \frac{60 \text{ min}}{1 \text{ hr}} \times \frac{1 \text{ L}}{1000 \text{ mL}} = 2.83 \text{ L/hr}$$

4. Calculate Total Volume of Distribution (Vd):

$$\text{Vd} = 7 \text{ L/kg} \times 70 \text{ kg} = 490 \text{ L}$$

5. Elimination Rate Constant:

$$k = \frac{\text{Cl}}{\text{Vd}}$$

$$k = \frac{2.83 \text{ L/hr}}{490 \text{ L}} \approx 0.0058 \text{ h}^{-1}$$

6. Half Life:

$$t_{1/2} = \frac{0.693}{k}$$

$$t_{1/2} = \frac{0.693}{0.00578 \text{ h}^{-1}} \approx 120 \text{ hours}$$

Question 98- A compounding pharmacist receives a prescription to prepare 100 mL of a 0.002% w/v solution of atropine sulfate. The pharmacy currently has pure atropine sulfate powder and a balance available. To ensure accuracy, the pharmacist decides to use the aliquot method. They will weigh the Minimum Weighable Quantity (MWQ) of the drug, mix it with lactose to form a stock mixture, and then weigh an aliquot from this mixture that is equal to the MWQ to obtain the exact amount of drug needed for the prescription. The sensitivity requirement is 6 mg and the maximum permissible error is 5%.

Calculate the amount of lactose diluent (in mg) that must be added to the weighed drug to create the stock mixture.

Question 98- Solution: 7080 mg

1. Calculate the amount of drug required for the prescription:

$$\% \text{ w/v} = \frac{\text{g solute}}{100 \text{ mL solution}}$$

$$0.002\% \text{ w/v} = 0.002 \text{ g per } 100 \text{ mL} = 2 \text{ mg.}$$

The prescription requires 2 mg of atropine sulfate.

2. Calculate the Minimum Weighable Quantity (MWQ):

$$\text{MWQ} = \frac{\text{Sensitivity}}{\text{Error}}$$

$$\text{MWQ} = \frac{6 \text{ mg}}{0.05} = 120 \text{ mg.}$$

3. Set up the Aliquot Calculation:

Since 2 mg is below the MWQ (120 mg), the aliquot method is necessary.

Drug Weighed: The pharmacist weighs the MWQ (120 mg) of the drug to satisfy the balance requirements.

Aliquot Weight: The pharmacist plans to weigh an aliquot equal to the MWQ (120 mg) from the final mixture.

Drug Needed: The aliquot must contain the 2 mg required for the solution.

4. Calculate the Total Stock Mixture Weight (W_{total}):

$$\frac{\text{Drug in Dilution}}{\text{Total Dilution Weight}} = \frac{\text{Drug in Aliquot}}{\text{Aliquot Weight}}$$

$$\frac{120 \text{ mg (Drug Weighed)}}{W_{\text{total}}} = \frac{2 \text{ mg (Drug Needed)}}{120 \text{ mg (Aliquot Weight)}}$$

Solving for W_{total} :

$$W_{\text{total}} = \frac{120 \text{ mg} \times 120 \text{ mg}}{2 \text{ mg}} = 7200 \text{ mg}$$

5. Calculate the amount of Diluent (Lactose) needed:

$$\text{Diluent} = W_{\text{total}} - \text{Drug Weighed}$$

$$\text{Diluent} = 7200 \text{ mg} - 120 \text{ mg} = 7080 \text{ mg}$$

Question 99- JL is a 64-year-old male patient (weight: 75 kg) admitted for the treatment of gram-negative sepsis. He received an initial IV bolus dose of 250 mg of gentamicin. Two plasma samples were drawn after the first dose, one 2 hours after and another 8 hours after. The plasma concentrations were 8.5 mg/L and 2.5 mg/L respectively.

Determine the patient's elimination rate constant (k), volume of distribution (V_d), and the predicted steady-state trough concentration ($C_{p_{\text{min,ss}}}$) if the 250 mg dose is continued every 12 hours.

Question 99- Solution: $k \approx 0.204 \text{ h}^{-1}$, $V_d \approx 19.6 \text{ L}$, $C_{p_{\text{min,ss}}} \approx 1.2 \text{ mg/L}$

1. Calculate Elimination Rate Constant (k):

$$k = \frac{\ln(C_{p1}) - \ln(C_{p2})}{t_2 - t_1}$$

$$k = \frac{\ln(8.5) - \ln(2.5)}{8 - 2} = \frac{2.140 - 0.916}{6} = \frac{1.224}{6} = 0.204 \text{ h}^{-1}$$

2. Calculate Initial Concentration (C_{p0}):

Using the concentration at 2 hours ($C_{p1} = 8.5 \text{ mg/L}$):

$$C_{p0} = \frac{C_{pt}}{e^{-k \times t}}$$

$$C_{p0} = \frac{8.5}{e^{-(0.204 \times 2)}} = \frac{8.5}{e^{-0.408}} = \frac{8.5}{0.665} = 12.8 \text{ mg/L}$$

3. Calculate Volume of Distribution (V_d):

$$V_d = \frac{D}{C_{p0}}$$

$$V_d = \frac{250 \text{ mg}}{12.8 \text{ mg/L}} = 19.5 \text{ L}$$

4. Calculate Steady State Peak ($C_{p_{\max,ss}}$):

Calculate the accumulation factor for $\tau = 12$:

$$C_{p_{\max,ss}} = \frac{D}{V_d \times (1 - e^{-k\tau})}$$

$$e^{-k\tau} = e^{-(0.204 \times 12)} = e^{-2.448} \approx 0.0865$$

$$C_{p_{\max,ss}} = \frac{250 \text{ mg}}{19.5 \text{ L} \times (1 - 0.0865)} = 14 \text{ mg/L}$$

5. Calculate Steady State Trough ($C_{p_{\min,ss}}$):

$$C_{p_{\min,ss}} = C_{p_{\max,ss}} \times e^{-k\tau}$$

$$C_{p_{\min,ss}} = 14 \text{ mg/L} \times 0.0865 \approx 1.21 \text{ mg/L}$$

Question 100- JT is a 34-year-old male patient with moderate persistent asthma who is prescribed an Advair HFA® (115 mcg fluticasone/21 mcg salmeterol) inhaler to manage his symptoms. Each inhaler device contains 120 metered actuations (puffs). The physician prescribes a total daily dose of 460 mcg of fluticasone and 84 mcg of salmeterol, to be divided equally and administered PO BID. The patient must release 4 actuations into the air to prime the device prior to their first use. Calculate the number of full days one inhaler canister will last for this patient.

Question 100- Solution: 29 days

1. Determine the daily puff requirement:

Calculate the total puffs needed per day using the fluticasone component:

$$\text{Total daily puffs} = \frac{460 \text{ mcg/day}}{115 \text{ mcg/puff}} = 4 \text{ puffs/day}$$

$$(\text{Verification with salmeterol: } \frac{84 \text{ mcg/day}}{21 \text{ mcg/puff}} = 4 \text{ puffs/day}).$$

Using the medical abbreviation reference, the instructions are PO BID (twice daily), which equates to 2 puffs/dose $\times$ 2 doses/day = 4 puffs/day.

2. Calculate the available actuations for therapeutic use:

The canister contains a total of 120 actuations.

The manufacturer instructions require the inhaler to be primed with 4 actuations prior to the first use, which cannot be used for the patient's dose.

Usable actuations = 120 puffs – 4 puffs = 116 puffs

3. Calculate the total days' supply:

Divide the total usable actuations by the daily puff requirement:

$$\text{Days Supply} = \frac{116 \text{ puffs}}{4 \text{ puffs/day}} = 29 \text{ days}$$

Change log**April 2026:**

- Question 34: Corrected a typo – 120 mg -> 120 mL

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